Algorithmic aspects of data-driven computational damage mechanics and introduction of Lipschitz regularization

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ABSTRACT

Data-driven computational mechanics enables the direct use of material databases in place of constitutive models, through an alternating minimization over mechanical and material states. This work investigates the behavior of this alternating minimization algorithm in the presence of damage-induced softening and snapback instabilities. By analytically prescribing the material response, database discretization effects are eliminated and the algorithmic behavior is isolated. The results show that tangent-consistent hyper-parameters and arc-length continuation are essential to achieve convergence and to follow post-peak response. A regularization strategy introducing an intrinsic length scale is then proposed and its effectiveness and implications are demonstrated through one- and two-dimensional numerical examples.

1. Introduction

The paradigm of Data-Driven Computational Mechanics (DDCM) was introduced in [21] as an alternative to classical constitutive modeling, replacing explicit material laws with a database of stress-strain pairs obtained, for instance, from experiments or multiscale simulations. In this framework, the solution of the boundary value problem is obtained by minimizing a distance functional defined between admissible mechanical states and material data. While the original work focused on linear elasticity, the approach has since been extended to inelastic problems, including plasticity and viscoelasticity [12], as well as to non-standard continua [20]. For higher-dimensional problems and for history-dependent materials, the application of DDCM requires specialized data structures and sampling strategies, which have been investigated in [11, 32].

Regularization plays a crucial role in DDCM, particularly when localization phenomena occur. In [20], DDCM was employed to predict the onset of shear bands in granular materials, and mesh-size dependence was mitigated by modeling the material as a Cosserat medium. This approach introduces additional kinematic and kinetic variables, together with a corresponding database for higher-order stresses and strains. While effective for shear-dominated problems, such regularization strategies are known to be inadequate for situations where shear is not the dominant deformation mode, such as mode-I crack propagation (see Section 7.3 of [34]). This limitation motivates the exploration of alternative regularization strategies within the DDCM framework, especially for fracture and damage problems.

The application of DDCM to fracture has been explored in several studies. In [8], fracture was modeled using a database relating energy release rate to crack length, with the crack evolution predicted through a data-driven minimization procedure; this approach was later extended to rate-dependent fracture in [7]. More recently, [35] combined DDCM with damage mechanics, introducing a micromorphic regularization to achieve mesh-independent predictions. In this setting, a length scale is incorporated through strain gradients, leading to the appearance of couple stresses, as commonly encountered in higher-order continuum theories [25, 36]. Consequently, the material database and the distance functional must be extended to include Cauchy stress-couple stress and strain-strain gradient pairs. This increases the dimensionality of the material phase space, which may lead to increased difficulties in sampling the database. The algorithmic behavior of such formulations in the presence of damage-induced instabilities remains largely unexplored.

Damage and fracture problems are often characterized by strong non-convexities and instabilities such as snapback. In conventional (non-data-driven) formulations, solution strategies typically exploit the separate convexity of the energy functional with respect to displacement and damage variables, enabling robust alternating minimization schemes. In such settings, displacement control is often sufficient to trace the equilibrium path, even in the presence of snapback. In contrast, the behavior of DDCM algorithms under damage-induced instabilities is far less understood. In the data-driven setting, the non-convexity of the strain energy is not expressed explicitly but is instead implicitly embedded in the material database. To date, investigations of instabilities in DDCM have largely focused on geometric nonlinearity. For instance, [22] studied a truss system exhibiting snap-through instability and analyzed the

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sensitivity of the DDCM solution path to perturbations and algorithmic parameters near limit points. However, corresponding studies addressing instabilities arising from material softening and damage are still lacking.

Another important aspect influencing the performance of DDCM algorithms is the choice of hyper-parameters, particularly the metric used in the distance functional. Previous works have shown that an appropriate choice of metric can significantly accelerate convergence. In [30], a locally linear reconstruction of the material database was used to define an effective local material response, while [26] demonstrated, using a truss example, that optimal convergence rates are obtained when the metric is chosen consistently with the tangent stiffness inferred from the database. These studies, however, focus on elastic-plastic or nonlinear elastic problems, for which the DDCM formulation remains well behaved. In such cases, the choice of metric primarily affects the rate of convergence. As will be shown in this work, the situation is fundamentally different in the presence of damage-induced non-convexity: here, the choice of hyper-parameters can critically influence not only the convergence rate but also whether the DDCM algorithm converges at all, and whether the obtained solution is physically meaningful.

Combining these considerations and building directly on [18], the objective of the present article can be established as follows. First, it is essential to analyze the alternating minimization procedure employed in the DDCM setting when softening effects are present. This, in turn, requires a suitable description of the material database incorporating softening behavior. Unlike [35], the present study assumes the availability of a database containing only classical stress-strain pairs, for instance obtained from experiments or lower-scale simulations that capture softening phenomena [29, 28, 27]. Regularization will be introduced separately from the material database. The database is assumed to be stored in a form that appropriately accounts for irreversibility effects; the specific data structures required for this purpose are not addressed here and can be found in [13, 11]. At the same time, it is well recognized that the performance of DDCM algorithms depends on the resolution of the material phase space, and that their sensitivity to database size remains largely unexplored in the presence of softening and instability phenomena. To isolate algorithmic effects from database discretization issues, throughout this article, the material response is therefore described implicitly through an underlying constitutive model, while the solution is still obtained by minimizing the data-driven distance functional with respect to the mechanical and the material states.

The article proceeds incrementally and the objectives of this work are:

- to investigate the convergence behavior of the DDCM alternating minimization algorithm in the presence of damage-induced softening and snapback instabilities, without any regularization in a simple setting;
- to analyze the influence of hyper-parameters on convergence and on the correctness of the obtained solution, without any regularization;
- to introduce within the DDCM framework a regularization strategy based on strain gradient limiters suitable for damage and fracture problems, first in 1D and then in 2D;
- to assess the proposed approach through a series of numerical examples involving damage and fracture.

The remainder of this article is organized as follows. First, the DDCM algorithm is briefly reviewed. A one-dimensional example is then introduced to analyze convergence properties in the presence of damage-induced non-convexity. Subsequently, the proposed regularization strategy is presented and extended to two-dimensional problems, followed by numerical examples illustrating the performance of the method.

2. Data driven solver

In this section, the data driven solver [21] will be introduced briefly. For this purpose, the notions of mechanical and material states are first introduced. A stress-strain pair, (σ, ϵ), constitutes a mechanical state if the stress state, σ, satisfies equilibrium and the strain state, ϵ, satisfies the compatibility, and the displacement fields satisfy the Dirichlet boundary conditions. At each point in the body, the material state, (σ^*, ϵ^*), belongs to the material manifold, $\mathcal{D}$, typically obtained from experiments or from simulations performed at lower scales. The equilibrium and compatibility manifold is defined as

$$\mathcal{E} = \{(\sigma, \epsilon) \mid \sigma, \epsilon \text{ are sufficiently smooth, } \operatorname{div}\sigma = 0 \text{ and } \epsilon = \operatorname{sym}\nabla u, u = u_D \text{ on } \partial\Omega_d\}. \tag{1}$$

The solution is sought as the intersection of the two manifolds.

$$\text{Sol} \in \mathcal{D} \cap \mathcal{E}. \tag{2}$$

In cases where the material behavior is defined by a discrete collection of stress-strain pairs, a strict intersection may not exist between the two manifolds. In such a case, the solution is sought as the mechanical-material states that are closest to each other with respect to a certain distance. In discrete case (such as in Finite Element Method, where each element in the mesh is associated with a pair of mechanical and material states), the distance is defined as

$$d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*) = \frac{1}{2}(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}^*)^T \boldsymbol{C}\boldsymbol{w}(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}^*) + \frac{1}{2}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^*)^T \boldsymbol{C}^{-1}\boldsymbol{w}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^*), \tag{3}$$

where $\boldsymbol{w}$ is vector containing the areas of elements, and the fourth-order tensor $\boldsymbol{C}$ is a hyper-parameter. $\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^* \in \mathbb{R}^{3N}$ in 2D, N denotes the number of elements, and $\boldsymbol{w} \in \mathbb{R}^{3N}$. $\boldsymbol{w}\boldsymbol{\sigma}$ thus indicates an element-wise operation. The discrete equilibrium manifold is defined as

$$\mathcal{E} = \{(\boldsymbol{\sigma}, \boldsymbol{\epsilon}) \quad | \boldsymbol{B}^T\boldsymbol{w}\boldsymbol{\sigma} = \boldsymbol{0} \text{ and } \boldsymbol{\epsilon} = \boldsymbol{B}\boldsymbol{u}, \boldsymbol{u} = \boldsymbol{u}_D \text{ on } \partial\Omega_d\}. \tag{4}$$

In the above, $\boldsymbol{B} \in \mathbb{R}^{3N\times 2N}$ denotes the discrete strain-displacement operator, $\boldsymbol{u} \in \mathbb{R}^{2N}$ is the displacement vector. Thus, the solution is the minimizer of the distance functional.

$$\text{Sol} \in \arg\min d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*) \text{ s.t. } (\boldsymbol{\sigma}, \boldsymbol{\epsilon}) \in \mathcal{E} \text{ and } (\boldsymbol{\sigma}^*, \boldsymbol{\epsilon}^*) \in \mathcal{D}. \tag{5}$$

The minimization can be carried out by imposing the constraints using Lagrange multipliers. The minimization process is divided into mechanical and material updates [21]. During the mechanical update, the material states, denoted by the asterisked quantities are held fixed. The minimization is carried out with respect to the mechanical states. During the material update, the mechanical states are held fixed and the minimization is carried out with respect to material states, typically element wise. The corresponding Euler Lagrange equations during the mechanical update can be seen to be:

$$\boldsymbol{B}^T\boldsymbol{C}\boldsymbol{w}\boldsymbol{B}\boldsymbol{u} = \boldsymbol{B}^T\boldsymbol{C}\boldsymbol{w}\boldsymbol{\epsilon}^*, \tag{6}$$

$$\boldsymbol{\sigma} - \boldsymbol{\sigma}^* = \boldsymbol{C}\boldsymbol{B}\boldsymbol{\eta}, \tag{7}$$

$$\boldsymbol{B}^T\boldsymbol{w}\boldsymbol{\sigma} = \boldsymbol{0}. \tag{8}$$

The last two of the above equations are combined and written as

$$\boldsymbol{B}^T\boldsymbol{w}\boldsymbol{C}\boldsymbol{B}\boldsymbol{\eta} = \boldsymbol{B}^T\boldsymbol{w}\boldsymbol{\sigma}^*. \tag{9}$$

$\boldsymbol{\eta}$ is treated as the perturbation on the displacement field and so is taken as $\boldsymbol{0}$ on the Dirichlet boundary. Equation (9) is used to obtain the remaining degrees of freedom.

The material update in the DD framework is performed at each Gauss point by searching for the point in the material database that is nearest to the current mechanical state. This step can be significantly optimized by employing specialized data structures for database storage and search, as discussed in [14]. The mechanical and the material updates are performed repeatedly until convergence. Refinement of the database ultimately leads to the convergence of the DD solution to the corresponding model-based solution [10].

Applications of DDCM to problems involving progressive damage and fracture are characterized by strong non-convexities arising from material softening. In conventional, model-based formulations, these non-convexities are typically handled through alternating minimization schemes that exploit the separate convexity of the energy functional with respect to displacement and damage fields, even in the presence of snapback. In contrast, the DD approach bypasses explicit constitutive modeling and embeds material nonlinearity implicitly through the material database. While it is known that DD solutions converge to their model-based counterparts in the limit of an infinitely rich database, the behavior of DD algorithms in the presence of damage-induced non-convexity and softening remains largely unexplored.

To investigate these effects in a controlled setting where the material phase space discretization effects are removed, the present work adopts a constitutive model-informed data-driven framework. Instead of relying on a finite and potentially poorly sampled database, the material update is performed using a known constitutive relation, which acts as an implicit, infinitely rich database. At each material point, the stress-strain pair is obtained through an analytical or numerical minimization of the data-driven distance functional subject to the constitutive relation. This approach is equivalent to a DD computation with an infinite dataset and eliminates errors associated with insufficient sampling of the material phase space. As a result, it allows the fundamental algorithmic challenges of applying DD mechanics based alternating minimization strategy to damage and fracture, particularly those related to non-convexity, softening, and convergence, to be isolated and examined.

Subsequently, a 1D problem without any regularization is considered first. The underlying constitutive model for the material database will allow a simple analysis of the 1D problem in various cases. Following this, the demonstration will be extended to the 2D case, with a regularization technique introduced later on.

3. Algorithmic robustness and convergence analysis

As mentioned earlier, the behavior of the alternate minimization (AM) algorithm employed in DDCM depends on the choice of hyper-parameters. More specifically, in the case of non-linear elasticity and plasticity, properly chosen hyper-parameters accelerate the rate of convergence of the algorithm [26, 30]. However, in the presence of softening, this section demonstrates that a proper choice of hyper-parameters is not only needed to achieve an optimal rate of convergence, but in the presence of instabilities (e.g., snapback), the choice, together with a continuation strategy, is critical even to achieve convergence.

Two properties are expected of the DDCM algorithm in this regime. First, in local damage models without a length scale, the damage localization region is mesh-dependent, shrinking to a zone of zero measure as the mesh is refined. The DDCM algorithm, when used with a 'local' database, must inherit this behavior; a diffuse damage zone in that setting would indicate a spurious algorithmic effect, not a physical one. Second, when a proper continuation strategy is employed, the evolution of the DDCM solution through softening-induced instabilities such as snapback can be tracked continuously. Both requirements need to be established before the algorithm can be employed in more general settings.

To isolate algorithmic effects from database sampling artifacts, the material database is replaced by the underlying constitutive model itself. Replacing a finite collection of data points by a material model is equivalent to considering a database with infinitely many, densely sampled points. This corresponds to a "best case scenario" for the DDCM solver and thus, any failure to converge to the correct solution in this setting necessarily implies failure with a finite database. Under this substitution, the material update at each Gauss point becomes

$$(\epsilon^*, \sigma^*) = \arg \min_{(\tilde{\epsilon^*}, \tilde{\sigma^*})} d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \tilde{\epsilon^*}, \tilde{\sigma^*}) \quad \text{s.t.} \quad \tilde{\sigma^*} = f(\tilde{\epsilon^*}; \mathcal{H}), \tag{10}$$

where f is the constitutive model relating material stresses and strains, and $\mathcal{H}$ is a history variable carried from the converged solution of the previous load step. The role of $\mathcal{H}$ is to encode irreversibility and to select the appropriate branch of the (multi-valued) softening response; its precise content, together with the branch-selection logic implemented by $f(\cdot; \mathcal{H})$, is given in Section 3.3.1 and in Algorithm 2 (Appendix A). The optimality condition then reads

$$\frac{\partial}{\partial \tilde{\epsilon^*}} \tilde{d}(\tilde{\epsilon^*}, f(\tilde{\epsilon^*}; \mathcal{H}); \boldsymbol{\epsilon}, \boldsymbol{\sigma}) = 0, \tag{11}$$

with $\tilde{d}(\tilde{\epsilon^*}, f(\tilde{\epsilon^*}; \mathcal{H}); \boldsymbol{\epsilon}, \boldsymbol{\sigma}) = d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \tilde{\epsilon^*}, \tilde{\sigma^*})$. The mechanical update is left unchanged. The overall alternating-minimization algorithm in this general setting is given as Algorithm 1 and discussed in Sections 3.2 and 3.3, where it is first exercised. In the analytical study of Section 3.1 that follows, $\mathcal{H}$ is held fixed within a load step through the branch restriction and the admissibility constraints (Definition 1), which isolates the fixed-point behavior of the alternating minimization on an already bifurcated branch.

In a 1D setting, if the material has linear softening, substituting a collection of data points with an implicit database further allows the successive iterates of the AM algorithm to be expressed as an affine map, and the fixed-point behavior of the DDCM iteration to be characterized analytically for the combinations of update operator and continuation strategy examined below. The constitutive model, serving here as an implicit database, is represented by a set of piecewise affine relations (see Figure 1): $\sigma = E\epsilon$ for the elastic regime, $\sigma = m\epsilon + c$ ($m < 0$) for the softening regime, and $\sigma = E_i\epsilon$ for the ith unloading branch (the unloading branches are not

considered in the analysis in Section 3.1). The material update is performed directly using the analytical expression of the constitutive model, as detailed in Appendix B.

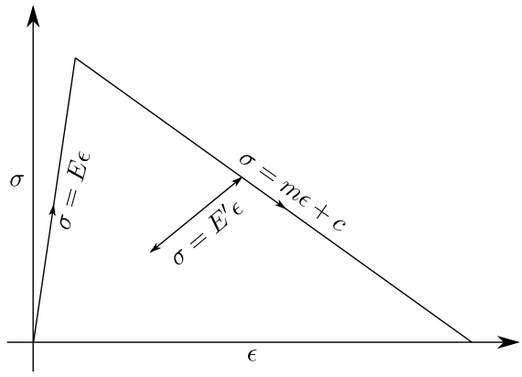


Figure 1: Stress strain database

The section proceeds as follows. First, a 1D bar is considered. To analyze the rate of convergence as well as the performance of the continuation strategy, the damage is taken to have already localized in an element - the solution path has thus already taken a non-homogeneous, bifurcated branch. The conditions required to continue on this branch are studied, establishing the ingredients that DDCM with softening depends on. These are then tested on a 1D bar subjected to a body force and on a bar with a notch in 2D.

3.1. Analysis of convergence in a 1D setting

Consider the case of a 1D bar of unit length, meshed with $N+1$ elements, where the element in the middle (without loss of generality) stays on the damage branch and the rest of the elements unload on the elastic branch. See Figure 2. Because the implicit local database carries no length scale, further loading concentrates straining in the middle element while the neighbors unload elastically. This section will analyze and establish the ingredients necessary for the DDCM solution to reproduce this behavior. Since the solution has already localized and is expected to remain so under further loading, elements on the elastic and damage branches are restricted to remain on those branches during the AM iterations. This restriction simplifies the analysis and makes it analytically tractable. Thus, the material state for the element on the damage branch is only searched along $\sigma^* = m\epsilon^* + c$, and for the elements on the elastic branch along $\sigma^* = E\epsilon^*$. This is referred to as *branch restriction* from hereon. The conditions under which the DDCM iterates continue along this bifurcated branch are studied below.

Under the branch restriction, the behavior of the AM iteration depends on two algorithmic choices: the hyper-parameters in the distance functional and the continuation strategy. If the material behavior is described by a local damage model (or sometimes even with a non-local model), a snapback is expected in the load displacement response of the body in some scenarios, depending on the mesh size, and the regularization length, when present. Two scenarios can thus be considered : one with a snapback and one without. In either case, the stress state is homogeneous in the entire bar as a consequence of equilibrium. The strain is localized in the element undergoing damage (in red in Figure 2), while the other elements (in blue) are unloading. Thus, the mechanical state of the bar is described by a scalar stress $\sigma \in \mathbb{R}$ and strains $\epsilon_1, \epsilon_2 \in \mathbb{R}$ (ϵ_1 for the undamaged elements and ϵ_2 for the damaged element). The corresponding material states are defined as $(\epsilon_1^*, \sigma_1^*) \in \mathbb{R}^2$ for the elements in blue and $(\epsilon_2^*, \sigma_2^*) \in \mathbb{R}^2$ for the element in red. The displacement at the right end of the bar is given as u_D and the left end is fixed. The distance function in this case is given by

$$d(\sigma, \epsilon_1^*, \epsilon_2^*, \epsilon_1, \epsilon_2; u_D) = \frac{N\ell_e}{2}\left[\frac{(\sigma-\sigma_1^*)^2}{C_1} + C_1(\epsilon_1-\epsilon_1^*)^2\right] + \frac{\ell_e}{2}\left[\frac{(\sigma-\sigma_2^*)^2}{C_2} + C_2(\epsilon_2-\epsilon_2^*)^2\right]. \tag{12}$$

The compatibility condition is $N\epsilon_1 + \epsilon_2 = u_D/\ell_e$, ℓ_e is the element length, C_1, C_2 are the hyper-parameters for the undamaged and damaged elements, respectively, and the number of elements is $N+1$. Introducing the compatibility condition into equation (12), Euler-Lagrange equations corresponding to the mechanical update can be obtained by setting the appropriate variations to 0. In the current case, Euler-Lagrange equations can be obtained for material updates as well by setting $d_{,\epsilon_1^*} = 0$ and $d_{,\epsilon_2^*} = 0$.

Definition 1 (Admissibility Constraints)**.** *To ensure that the material states obtained during the AM iterations are admissible, the following constraints are placed on the material iterates.*

1. *For the element on the damage branch, if the damage state at the beginning of the load step is d_0, any material iterate whose associated damage falls below d_0 is discarded and is replaced by the material state carried over from the previous load step.*

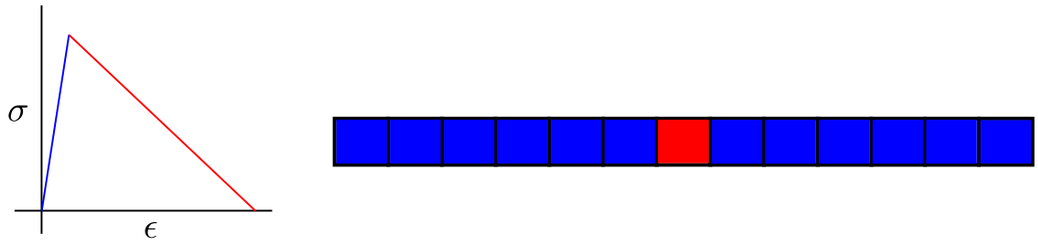


Figure 2: 1D bar with damaged element and the associated elastic (in blue) and damage (in red) branches.

2. *If the material state iterate for an elastic element yields $\sigma^* > \sigma_{\max}$, where $\sigma_{\max}$ is the peak stress of the implicit database, the state is replaced by* $(\sigma_{\max}/E,\ \sigma_{\max})$.

Defining $\boldsymbol{a} = [\epsilon_1, \epsilon_2, \sigma]^T$, $\boldsymbol{b} = [\epsilon_1^*, \epsilon_2^*, \sigma_1^*, \sigma_2^*]^T$, as shown in Appendix C, the mechanical and the material updates within a load increment can be written concisely as

$$\boldsymbol{a}^{i+1} = \boldsymbol{A}\boldsymbol{b}^{i+1} + \boldsymbol{c}, \tag{13}$$

$$\boldsymbol{b}^{i+1} = \boldsymbol{C}\boldsymbol{a}^{i} + \boldsymbol{d}, \tag{14}$$

where $\boldsymbol{A}, \boldsymbol{C}, \boldsymbol{c}, \boldsymbol{d}$ are parameters defined in Appendix C. i denotes the iteration number. The updates can be combined and written together as

$$\boldsymbol{a}^{i+1} = \boldsymbol{A}\boldsymbol{C}\boldsymbol{a}^{i} + \boldsymbol{f}. \tag{15}$$

At convergence, $\boldsymbol{a}^{i+1} \approx \boldsymbol{a}^{i} \approx \boldsymbol{a}$, where $\boldsymbol{a}$ is the exact solution. Defining the error at step i as $\boldsymbol{e}^i = \boldsymbol{a}^i - \boldsymbol{a}$, the errors at successive iterations can be related, using equation 15, as

$$\boldsymbol{e}^{i+1} = \boldsymbol{A}\boldsymbol{C}\boldsymbol{e}^{i}. \tag{16}$$

Thus the rate of convergence of the method (and whether it converges at all) can be determined by analyzing the matrix $\boldsymbol{AC}$, that corresponds to the combined mechanical and the material updates. As reported in Appendix C, the iterates can be cast into the form in equation 15 even when using an arc-length solver, with the applied displacement u_D now added to the mechanical state as an unknown.

With this framework, the 1D bar in Figure 2 will now be analyzed for various cases as stated below.

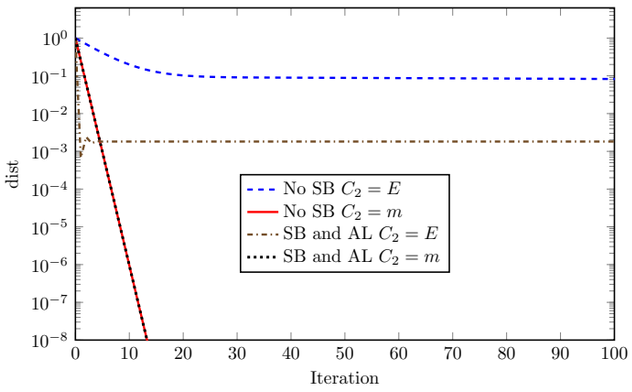
1. *The effect of the hyper-parameters on convergence rate for continuous evolution.* For this, a displacement-driven system will be considered without snapback, examining two choices of C_2.
2. *The effect of the hyper-parameters under snapback.* For this, arc-length solver will be used, examining the same two choices of C_2.
3. *The effect of the driving mechanism under snapback.* For this, a displacement controlled system will be used, with two choices of C_2.

It will be assumed that the solution at some load step n is known. The external load u_D is then varied. The intent is to find the solution to the 1D problem, given the state at step n, under the new loading condition. The evolution of the mechanical and the material states through the DDCM iterations will be examined in two ways. First, using the distance function in equation 12 with $C_1 = C_2 = E$. Thus,

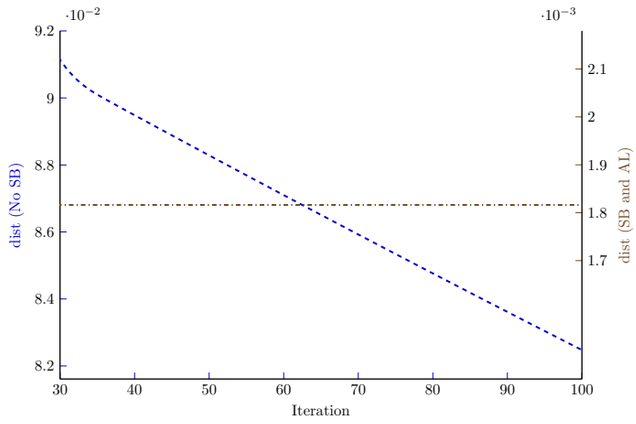
$$d_{\text{conv}}(\sigma, \sigma_1^*, \sigma_2^*, \epsilon_1, \epsilon_2; u_D) = \frac{N\ell_e}{2}\left[\frac{(\sigma-\sigma_1^*)^2}{E} + E(\epsilon_1-\epsilon_1^*)^2\right] + \frac{\ell_e}{2}\left[\frac{(\sigma-\sigma_2^*)^2}{E} + E(\epsilon_2-\epsilon_2^*)^2\right]. \tag{17}$$

Note that the parameters are set to E only to evaluate the distance and not to compute the DDCM iterates. Second, the trajectory of the mechanical and the material states followed by the elements on corresponding branches within the mechanical-material minimization process of the DD algorithm will be presented on a $\sigma - \epsilon$ space.

On the trajectory plot, the mechanical and the material states of the intermediate iterates separate two groups: those corresponding to the elements on the elastic branch and the one corresponding to the element on the damage branch. The evolution of the AM iterates can thus be monitored by tracking the states of each group. In line with the color scheme presented in Figure 2, the states on the elastic



(a) Evolution of the normalized distance with iterations, on log scale.



(b) Linear zoom of the $C_2 = E$ cases over iterations 30–100. Note the different scales on the two axes: the plot compares slopes (contraction vs. non-contraction).

Figure 3: Evolution of the distance functional d_{conv} (equation 17), normalized by its value at the first iteration, for the cases 1 and 2 studied in Section 3.1. No SB: displacement control with $N = 10$ (no snapback). SB and AL: arc-length control with $N = 100$ (snapback regime).

branch are marked in blue, while those on the damage branch are marked in red. The associated material states are marked using circles $\circ$, and the mechanical states using crosses $\times$, and the initial iterate on each branch is marked with a black asterisk $*$. Together with the evolution of the distance functional, this presents a complete picture of the behavior of the DDCM solver under the three cases stated earlier.

For all the cases considered below, the material parameters are taken as $E = 100\sigma_0, m = -7.1428\sigma_0, c = 107.142\sigma_0$, where σ_0 is a reference stress and all stresses in this section are expressed in units of σ_0. The resulting stress-strain response has a maximum stress $\sigma_{\max} = 100\sigma_0$, strain at damage initiation $\epsilon_i = 1$, and strain at complete failure $\epsilon_f = 15$. The initial mechanical and material states are taken as $\sigma = \sigma_1^* = \sigma_2^* = 99.95\sigma_0, \epsilon_1 = \epsilon_1^* = 0.9995, \epsilon_2 = \epsilon_2^* = 1.007$. This state corresponds to points that are just below the cusp of the damage initiation point. The following analysis is performed with the *branch restriction* and the admissibility constraints defined above. These restrictions are introduced solely to isolate the fixed-point properties of the DD alternating minimization near a given bifurcated solution and do not reflect an assumption of the general DD algorithm. In particular, branch restriction is removed in subsequent examples, where the method is allowed to select material states freely (albeit under admissibility constraints).

Behavior without snapback under displacement control

To isolate the effect of the hyper-parameters on the convergence rate corresponding to case 1, this section examines the behavior of the DDCM AM algorithm under displacement control when a snapback instability does not take place. For the setup under consideration, when $mN + E > 0$, a snapback does not occur and the solution evolution is smooth, and can be captured by direct displacement control. $mN + E > 0 \implies N < -E/m = 14$. So, $N = 10$ will be considered. The new applied displacement is taken as $u_D = 1.0104$.

The analysis begins by taking C_1, C_2 equal to E. The distance function is normalized with its value at the first iteration and its evolution is presented in Figure 3a using a blue dashed line. The distance initially decreases and appears to settle at a certain value. However, a close up between the iterations 30–100 presented in Figure 3b indicates that the distance still decreases, albeit at a very slow rate. To investigate further, the intermediate iterates of the mechanical and the material states, $(\epsilon_1, \sigma), (\epsilon_2, \sigma)$ and $(\epsilon_1^*, \sigma_1^*), (\epsilon_2^*, \sigma_2^*)$, respectively, for this case, computed from equations (14) and (13), are presented in Figure 4a on a $\epsilon - \sigma$ plot. At the first iteration, the mechanical states for all the elements move towards the damage branch. During the subsequent iterations, the mechanical states corresponding to the elements on the elastic branch return back, while the state of the element on the softening branch continues to go further along the branch. With increasing iteration count, the iterates on the elastic branch become progressively closer to one another, and the same behavior is observed for the damage branch. At some point, successive iterates can no longer be distinguished in Figure 4a, even though the mechanical and the material states have not coincided and the distance functional remains bounded away from zero. This is in line with the observations regarding the distance functional reported in Figures 3a and 3b.

The reason behind this observation can be obtained by analyzing the matrix $\boldsymbol{AC}$ in equation (15). The matrix $\boldsymbol{AC}$ is not symmetric. But for the parameters listed above, its eigenvalues are real. Thus, the rate of convergence of the iterations in equation (15) can be obtained from the spectral radius of the matrix $\boldsymbol{AC}$. When C_1, C_2 are taken equal to E in equations (57) and (58), the spectral radius

of $\boldsymbol{AC}$ is 0.9994. This implies that the fixed point iterations in equation (15) are contractive and hence converge, albeit at a very slow rate. This explains the closeness of the successive iterates in Figure 4a and the slowness of the decay in the distance reported in Figures 3a and 3b. If a discrete database were used instead of the analytical expressions, an extremely fine database would be required to ensure that the distance functional reduces sufficiently at convergence. In practice, DDCM implementations declare convergence when successive iterates cease to move, since with a sparse database the distance functional cannot in general be driven to zero. Under $C_2 = E$, however, the iterates move imperceptibly between AM steps long before the mechanical-material gap has sufficiently reduced; the near-stationarity of the iterates would then spuriously imply convergence while the distance remains large. Diagnosing this pathology in practice requires either an implicit database, as employed here, or a discrete database sampled finely enough that the material projection follows the mechanical state closely between iterations.

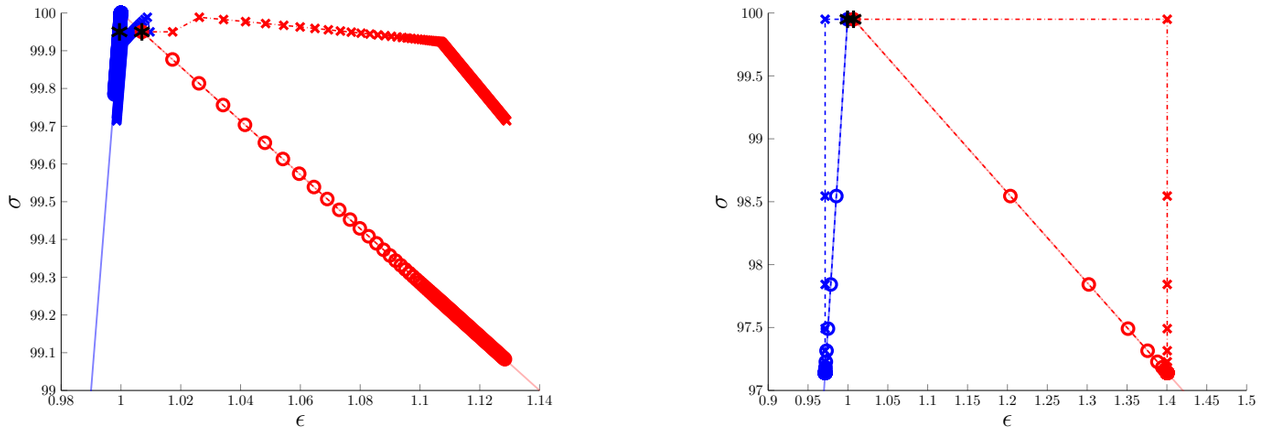
Following this, the scenario $C_2 = m$, which corresponds to the derivative $\partial\sigma^*/\partial\epsilon^*$, is examined to assess how this choice affects the rate of convergence. Since $m < 0$, this formally corresponds to assigning a negative weight to the corresponding terms in the distance functional, which would be inadmissible if used to define the metric itself. For this reason, $C_2 = m$ is employed only in the Euler-Lagrange equations (57) and (58). The choice $C_2 = m$ therefore does not alter the underlying metric and should be interpreted solely as an algorithmic modification used to probe its effect on convergence behavior. In the Euler-Lagrange equations, the choice $C_2 = m$ results in a left-hand side that corresponds to a consistent linearization of the condensed energy density, since $m = \partial\sigma^*/\partial\epsilon^*$. The positive-definite distance interpretation and the associated variational structure are lost, but the operator now reflects the indefiniteness intrinsic to softening.

The matrix $\boldsymbol{AC}$ remains unsymmetric with $C_2 = m$, but still possesses real eigenvalues. Its spectral radius in this case is 0.5. The evolution of normalized distance in this case is reported in Figure 3a using a solid red line. The distance decays rapidly and reaches a value of about 1×10^{-8} in about 10 iterations. The observed decay rate is consistent with the spectral radius $\rho = 1/2$ derived analytically. The (deceptive) plateau observed for $C_2 = E$ is no longer present. To examine this in more details, the successive iterates of mechanical and material states using equations (13) and (14) are presented in Figure 4b. Unlike the behavior observed in Figure 4a, the iterates are more widely spread. The mechanical and material states approach each other over successive iterations and tend to intersect at a rate much faster than when $C_2 = E$. The solution converges rapidly. Also, when $C_2 = E$, during the first iteration, the (mechanical) strain changes for the elements on the elastic and the damage branch point in the same direction (positive) as the external load (also positive). In contrast, when $C_2 = m$, the sign of the strain change for the element on the damage branch remains the same as the external load, while for the element on the elastic branch, it is opposite.

Remark. *When $C_2 = E$, the initial mechanical strain increment for the elements on the elastic and softening branch are in the same direction. If the elements are not restricted to lie on their corresponding branches during the course of iterations, the system will ~~thus~~ favor the homogeneous branch of the solution, since starting from iteration 2, the nearest points for the material state of the elements in blue lie on the damage branch. See Figure 4a. Choosing $C_2 = m$ will let the system continue on the existing bifurcation branch as presented in Figure 4b. Taking $C_2 = E$ is thus likely to cause the solution to revert from the bifurcated path to the homogeneous one, an outcome that is undesirable when the objective is to capture localized damage evolution. This will be of importance in 2D, as shown later.*

Behavior with snapback and arc-length solver

When $N > -E/m = 14$, a snapback instability takes place. Thus, to isolate the effect of the hyper-parameters under snapback corresponding to case 2, $N = 100$ is considered in this section. Typically, the alternating minimization between displacement and damage variables in model-based simulations with softening is capable of navigating the snapback, albeit needing a large number of iterations to converge. The solution typically 'jumps through' the snapback under displacement control. On the other hand, arc-length solvers can also be employed to follow the gradual evolution of damage. This scenario will be examined using the DDCM algorithm on the example described above, but using an arc length solver to follow the smooth evolution of the solution. The material parameters and the initial conditions are kept the same as in the previous case, but with $N = 100$. The solution update using an arc length solver is presented in equations (61) and (60). The arc length constraint corresponds to prescribing the strain value for the element on the damage branch, in the current case as 1.4. Thus, $\epsilon_2 = \epsilon_0 + \Delta\epsilon = 1.4$ in equation (60). The prescribed displacement u_D is obtained as a part of the solution.



(a) The paths taken by the mechanical and the material states for $C_2 = E$.

(b) The paths taken by the mechanical and the material states for $C_2 = m$.

Figure 4: Transition of the mechanical and the material states. Case 1: without snap back, using displacement control. In both panels, mechanical states are shown as crosses (×) and material states as circles (∘); the initial iterate on each branch is marked with a black asterisk (∗). The underlying elastic and damage branches are indicated using the same color scheme as in Figure 2: blue for the elastic branch, red for the damage branch.

The evolution of the distance functional when $C_2 = E$ is presented in Figure 3a using a dot-dashed gray line, with a closeup view in Figure 3b. After an initial decrease, the normalized distance saturates at a value of about 10^{-3} and remains there for the remaining iterations. The close-up in Figure 3b confirms this saturation: unlike the $C_2 = E$ curve of the no-snapback case, which continues to decay slowly, this one has stopped moving entirely. The origin of this behavior is the spectral radius of the composed update operator: for $N = 100$ and $C_2 = E$, $\rho(\boldsymbol{AC}) = 1.029 > 1$. The fixed-point iteration is non-contractive, and no amount of iteration will close the distance between mechanical and material states. Yet $\rho(\boldsymbol{AC}) > 1$ does not translate to divergence - the iterates freeze at a finite distance. This is the consequence of the constraint 2: the material projection saturates at the peak stress. The iterates leave the regime governed by $\boldsymbol{AC}$; this constraint-saturation mechanism is detailed in the following subsection.

The trajectory of the iterates in Figure 5a makes the mechanism visible. The arc length constraint carries all the mechanical strains near the prescribed value of 1.4 at the first iteration. Under branch restriction, the element on the damage branch then remains where it landed after the first iteration, while the states of the elements restricted to the elastic branch return. If the branch restriction is removed, the failure takes a different form: from the second iteration onward, the nearest admissible material state for the elements initially on the elastic branch is located on the damage branch, and the solution migrates to the homogeneous, unlocalized configuration, even when the initial iterate lies slightly away from the elastic-damage cusp. Similar conclusions about the convergence can be drawn even if the starting point is further down the damage branch. This case has been presented in appendix D.

When $C_2 = m$, the spectral radius is $\rho(\boldsymbol{AC}) = 0.5$. Correspondingly, as presented in Figure 3a using a dotted black line, the normalized distance decays rapidly, reaching a value below 10^{-8} in about 10 iterations. To investigate further, the paths taken by the successive iterates are presented in Figure 5b. The strain increments for the elements on the elastic and the softening branches point in opposite directions, unlike the case when $C_2 = E$. The mechanical and the material states quickly reach the intersection point. This stresses the importance of the hyper-parameters even when driven using an arc-length constraint, specifically in the case where a snapback is expected to occur. In this case, it is necessary to take $C_2 = m$ to arrive at a converged solution.

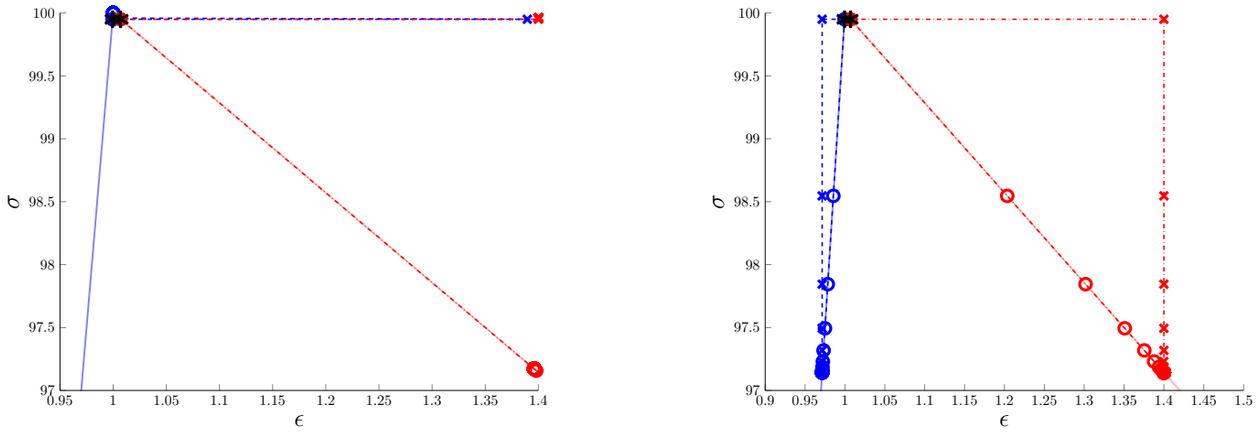
The results presented thus far using an arc-length solver can be formalized by examining the effect of the parameter C_2 on the spectral radius of the iteration matrix $\boldsymbol{AC}$. Using the expressions presented in Appendix C for the case with arc-length solver, the characteristic polynomial of the matrix $\boldsymbol{AC}$, after eliminating the double root of 0, can be derived as

$$\lambda^2 - T\lambda + \Delta = 0, \tag{18}$$

where, if $C_1 = E$,

$$T(C_2, N) = 1 + \frac{E\,(m^2 - C_2^2)}{2\,D\,(m^2 + C_2^2)}, \tag{19}$$

$$\Delta(C_2, N) = \frac{m\,(NC_2^2 + Em)}{2\,D\,(m^2 + C_2^2)}. \tag{20}$$



(a) The paths taken by the mechanical and the material states for $C_2 = E$.

(b) The paths taken by the mechanical and the material states for $C_2 = m$.

Figure 5: Transition of the mechanical and the material states. Case 2: with snap back, using arc-length.

$D = C_1 + NC_2$. Thus, the spectral radius is $\rho(C_2, N) = \max(|\lambda_+|, |\lambda_-|)$ with $\lambda_\pm = T/2 \pm \sqrt{T^2/4 - \Delta}$. The case $C_2 = m$ recovers the value observed above: $T = 1$, $\Delta = 1/4$, and $\rho = 1/2$.

It remains to determine the range of C_2 for which the iteration is contractive, i.e., $\rho < 1$. Tt can be verified that $T^2 \geq 4\Delta$, so both eigenvalues are real. Thus, λ_+ is smaller than 1 if $\Delta > T - 1$, which translates to

$$\frac{E + Nm}{E + NC_2} > 0. \tag{21}$$

The snapback regime corresponds to $N > -E/m$, in which the numerator of (21) is negative. Contraction of the AM iterations under snapback therefore requires the denominator to be negative as well, giving the necessary condition

$$C_2 < -E/N. \tag{22}$$

Under snapback, the AM fixed-point map is contractive only for C_2 strictly negative. Since $E > 0$ and $N > 0$, no positive value of C_2 satisfies (22). The tangent-consistent choice $C_2 = m$ lies in the admissible range because for a snapback to take place, $N > -E/m \implies m < -E/N$ and thus $C_2 = m < -E/N$; it is the unique choice that is simultaneously an interpretable derivative of the constitutive response and an admissible hyper-parameter for contraction.

The analysis can also be used to establish the conditions for convergence in the absence of snapback as in case 1, when the system is driven by an arc-length solver. In the absence of snapback, $N < -E/m$, and so $E + Nm > 0$. Convergence therefore requires $E + NC_2 > 0$, that is $C_2 > -E/N$. In this case, both choices $C_2 = m > -E/N$ and $C_2 = E > -E/N$ result in convergence, albeit at varying rates, with the choice $C_2 = E$ resulting in a spectral radius close to 1.

Failure with snapback and displacement control

To assess whether the arc-length solver is strictly necessary, i.e., whether the data-driven algorithm can intrinsically traverse snapback instabilities under pure displacement control, the snapback case is examined with the end displacement u_D prescribed directly, without arc-length continuation, which corresponds to case 3 listed earlier. $N = 100 > -E/m$ to ensure a snapback takes place. The evolution of normalized distance for the two scenarios $C_2 = E, m$ is presented in Figure 6 using a dashed blue line and a solid red line, respectively. In both the cases, the distance stagnates at values greater than 0.1. The corresponding DD iterations are shown in Figures 7a and 7b.

Before discussing the results, it is important to delimit the range of validity of the spectral analysis of Section 3.1. The estimate based on the spectral radius of $\boldsymbol{AC}$ presumes that the iteration coincides with the affine map $\boldsymbol{a}^{i+1} = \boldsymbol{AC}\,\boldsymbol{a}^i + \boldsymbol{f}$. This holds only while every material projection returns a state lying on the corresponding branch and is admissible, so that the closed-form updates of Appendix C apply unmodified. It is possible that the intermediate material states obtained from equation (14) are inadmissible - that is, they may activate either or both of the admissibility constraints 1 and 2. When either constraint becomes active, the material update is no longer the linear operator $\boldsymbol{A}$, but $\boldsymbol{A}$ composed with a projection onto the admissible segment. In that regime the iterated

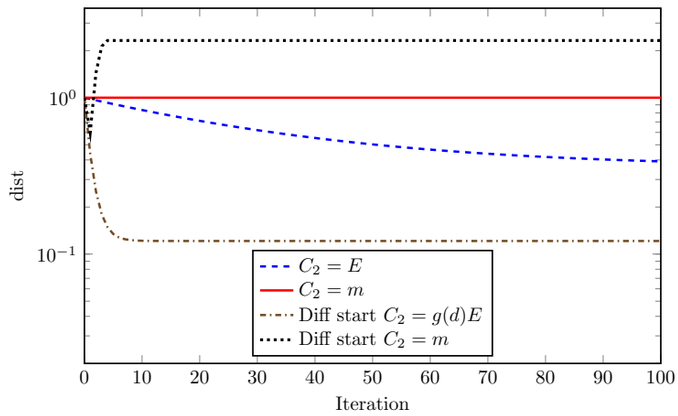


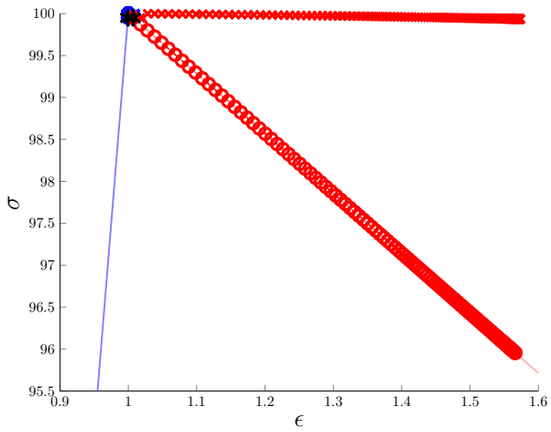
Figure 6: Evolution of the normalized distance functional d_{conv} under displacement control in the snapback regime ($N = 100$), case 3. Main starting point (near cusp): $C_2 = E$ (dashed blue) stalls near 4×10^{-1}; $C_2 = m$ (solid red) barely moves from its initial value. Different starting point further down the softening branch (see Appendix D): $C_2 = g(d)E$ (dot-dashed brown) drops and plateaus near 10^{-1}; $C_2 = m$ (dotted black) diverges and plateaus above 10^0. All four curves stagnate at finite distances, indicating that the AM iteration fails to reduce the mechanical-material gap under displacement control, independent of the hyper-parameter and the initial state.

Table 1
Summary of the three cases examined in Section 3.1. Only the tangent-consistent metric combined with arc-length continuation recovers the localized bifurcated solution under snapback.

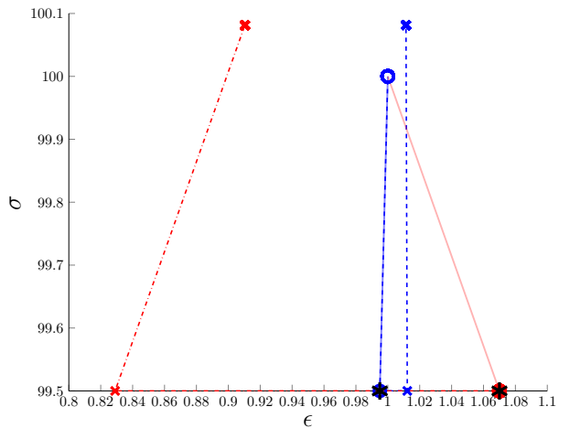
Case	Regime	$C_2 = E$ (or $g(d)E$)	$C_2 = m$
1	No snapback, disp. ctrl.	Converges ($\rho \approx 0.9994$)	Converges ($\rho = 1/2$)
2	Snapback, arc-length	**Fails**	**Converges** ($\rho = 1/2$)
3	Snapback, disp. ctrl.	Fails	Fails

map is no longer $\boldsymbol{AC}$, and its spectral radius, however computed for the interior map, no longer governs the behavior of the iteration. This is precisely what occurs under displacement control beyond the snapback threshold.

When the initial state lies near the elastic-damage cusp, the prescribed u_D drives the mechanical update toward stresses exceeding σ_{peak}. The database contains no such states, so the material projection saturates at the peak as a consequence of constraint 2. The mechanical and material states then cannot coincide: the equilibrium-compatibility manifold and the material manifold do not intersect in the reachable region for the prescribed u_D. The distance functional consequently plateaus at, or grows to, a finite value and never closes, irrespective of the spectral radius of the interior map. This behavior is shown for $C_2 = E$ in Figure 7a. If the branch restriction was removed, the solution again moves to the homogeneous branch. Similar results are observed as presented in Figure 7b even with $C_2 = m$. The scenario where the initial condition is further below the yield cusp, presented in appendix D, reveals similar results. Thus, in the presence of snapback, displacement control is unable to track the solution path and an arc-length continuation strategy, with an appropriate choice of hyper-parameters is necessary.



(a) The paths taken by the mechanical and the material states for $C_2 = E$.



(b) The paths taken by the mechanical and the material states for $C_2 = m$.

Figure 7: Transition of the mechanical and the material states. Case 3: with snap back, using displacement control.

Summary

The three cases examined in this subsection are summarized below in table 1. From the analyses, it can be concluded that a proper choice of hyper-parameters together with a proper continuation strategy is necessary for the AM iterates in DDCM to converge to a correctly localized solution in the presence of softening. In the presence of instabilities such as snapback, the tangent operator ceases to be positive definite and a proper continuation technique is necessary to trace the equilibrium solution path [6, 5, 3].

In the case of DDCM where the material behavior is stable, for instance in the case of a non-linearly elastic material, an optimal value for C_2 is only used to accelerate the convergence [26]. In the presence of instabilities, structural or material, the scenario is different. For instance, the successive mechanical and material iterates presented earlier resemble the path taken by the iterates in a truss system, presented in Figures 6, 7, and 16 of [22]. In that reference, the performance of DD solver has been analyzed for cases involving instabilities. A truss system was considered, with a linearly elastic material behavior. The authors of [22] extract the eigenvector corresponding to the zero eigenvalue of the hessian matrix to perturb the initial condition in DD solution. They also reverse their loading when the hessian matrix develops a zero eigenvalue. This is thus similar to implicitly using an arc-length solver. In the current case, the origins of the instability are both material (softening) and geometric (mesh size/ characteristic length). Thus, an arc-length solver will be used to trace any and all the snapback instabilities.

In the present work, the underlying material model is characterized by linear softening, which allows the convergence behavior to be analyzed through the spectral radius of the matrix $\boldsymbol{AC}$. If this condition is not satisfied, for example, if the softening response is nonlinear or the problem is formulated in two dimensions, such an analysis is no longer feasible, since $\boldsymbol{AC}$ then depends on the evolving strain state. Nevertheless, the two-dimensional analyses presented later consistently converge in approximately 25 iterations per load step when the material tangent is used as the hyper-parameter, in line with the fast convergence observed in the 1D cases.

The analysis performed so far restricts the elements on the elastic and damage branches to remain there. A continuation strategy with an appropriate (tangent) hyper-parameter choice is necessary to obtain a solution that is localized. The following section removes the branch restrictions and examines whether the arc-length plus tangent-hyper-parameter combination reproduces the mesh dependence expected from a local implicit material database. For this purpose, a 1D bar subjected to a body force will first be studied. This will be followed by a 2D example - a bar with a notch.

3.2. A 1D bar with a body force

In contrast to the analytical study of Section 3.1, no branch restriction is imposed in the following example. The material update is allowed to select the material state at each iteration as described below, and the observed behavior therefore reflects the intrinsic branch-selection properties of the DD algorithm. The solution evolution still has to satisfy the irreversibility of damage: the material cannot heal. Also, when the implicit database consists of elastic and softening branches (see Figure 1) it is necessary to determine the appropriate solution branch depending on the current loading and the solution history. The material update phase of the AM iterations, considering these requirements is presented in Algorithm 2. A condensed version for 1D case is presented below and will be followed by a detailed description in the 2D case that will follow. The overall mechanical and material updates follow the structure presented in algorithm 1.

Material update

At each Gauss point, the converged solution at load step n carries a damage variable D^n and a strain-history variable $\epsilon^n_{\max} = \max_{k\leq n} \epsilon^k$. To be noted is that these are not independent - D^n can be evaluated using $\epsilon^n_{\max}$. These are the state variables the material update reads at iteration i of load step $n+1$.

The constitutive model, detailed in Appendix B, is the linear-softening damage law $\sigma = g(D)E\epsilon$, with degradation function

$$g(D) = \frac{1-D}{1+(k-1)D}, \qquad k = \epsilon_f/\epsilon_i, \tag{23}$$

where ϵ_i is the strain at damage initiation and ϵ_f is the strain at $D = 1$. The damage evolution is

$$D = \frac{\epsilon_{\max} - \epsilon_i}{\epsilon_f - \epsilon_i}, \qquad \epsilon_{\max} \geq \epsilon_i. \tag{24}$$

The piecewise-affine model of Section 3.1 is recovered with $m = E\epsilon_i/(\epsilon_i - \epsilon_f)$ and $c = -m\epsilon_f$. At iteration i of load step $n+1$, given the current mechanical state (ϵ, σ) at a Gauss point, the material update returns the appropriate state using the methodology in section 3.3.1. At convergence of the AM iterations for load step $n+1$, the state variables are updated: $D^{n+1} = D(\epsilon^{n+1}_{\max})$ and $\epsilon^{n+1}_{\max} = \max(\epsilon^n_{\max}, \epsilon^{n+1})$.

Mechanical update

The mechanical update is unchanged from Section 3.1 except for the arc-length constraint (Appendix E) and the element-wise assignment of $\boldsymbol{C}$ from the material tangent. Expressing the displacement vector as $\boldsymbol{u} = \boldsymbol{C}_f \boldsymbol{u}_f + \lambda_g \boldsymbol{u}_g$, where $\boldsymbol{u}_f$ are the unknown degrees of freedom, λ_g is the strength of the applied boundary condition, $\boldsymbol{u}_g$ corresponds to the Dirichlet degrees of freedom, and $\boldsymbol{C}_f$ is a matrix of zeros and ones that relates $\boldsymbol{u}_f$ to $\boldsymbol{u}$; similarly $\boldsymbol{\eta} = \boldsymbol{C}_f \boldsymbol{\eta}_f$. Under displacement control, λ_g is known and equations (6) and (9) are solved for $\boldsymbol{u}_f$ and $\boldsymbol{\eta}_f$. Under arc-length control, λ_g is treated as an additional unknown and the system is supplemented with an arc-length constraint (Appendix E).

The problem of a 1D bar, 1 m long, subjected to a unit body force will now be analyzed. The left end is fixed and a displacement is prescribed at the right end; the body force acts uniformly along the length of the bar. The right-end displacement is treated as an unknown and driven by arc-length continuation on the strain of the leftmost element (Appendix E). The bar is discretized with piecewise-affine continuous displacement elements and piecewise-constant stresses over a homogeneous mesh. The two mesh sizes, ℓ_e and $0.5\ell_e$, are considered to probe mesh sensitivity. The material parameters are those of Section 3.1, and the implicit database is illustrated in Figure 1. Arc-length control is used with the tangent-consistent hyper-parameter assignment established in Section 3.1. The same problem will be revisited in Section 4.1 to demonstrate the effect of introducing a length scale.

Algorithm 1 The DD algorithm

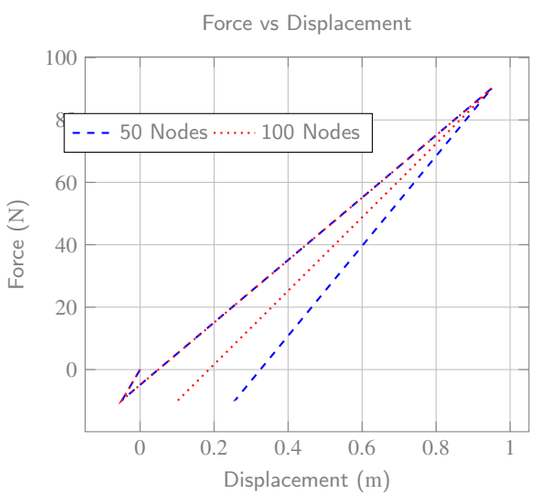
```
Given σ^n, ε^n, σ*^n, ε*^n, Δε
Determine σ^{n+1}, ε^{n+1}, σ*^{n+1}, ε*^{n+1}
i ← 1, σ*_0 = σ*^n, ε*_0 = ε*^n
while not converged do
    procedure MECHANICAL UPDATE(σ*_{i−1}, ε*_{i−1}, C)
        Resolve the Euler–Lagrange equations
        return σ_i, ε_i
    end procedure
    procedure MATERIAL UPDATE(σ_i, ε_i, D^n, H^n)
        Apply Algorithm 2
        return (ε*_i, σ*_i, C_i, D_i)
    end procedure
    i ← i + 1
end while
n ← n + 1
```

The force at the right end of the bar vs the applied displacement until complete failure (where some element in the bar is fully damaged) for the two mesh sizes has been presented in Figure 8a. The mesh dependence of the strain distribution, together with the snapback response, can be clearly seen once the *damage begins to increase*. At the first step, a compressive force is developed as a consequence of the body force, which gradually becomes positive as the displacement is increased. The initial negative displacement is a consequence of the arc-length constraint that the strain increment in the left most element is equal to a certain (user specified) value. As the load is increased further, the damage initiates at the left end (solely at the left end due to the applied body force). The strain and the damage localize in the element at the left end in the subsequent steps until the element undergoes a complete failure ($d = 1$). A small compressive load can again be seen at the end as a consequence of the applied body force. The corresponding strain distributions, at complete failure, for the two mesh sizes can be seen in Figure 8b. The mesh dependence of the solution is evident. Using appropriate values for the hyper-parameters, together with the arc-length solver accurately captures the snapback response of the bar, mesh dependence of the solution, and follows the expected bifurcation branch.

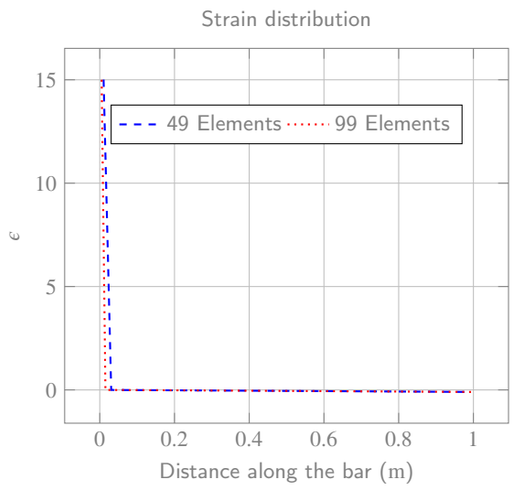
The 1D case examined so far uses a linear softening model in which the tangent $m = \partial\sigma^*/\partial\epsilon^*$ on the damage branch is constant. This constancy is what made the closed-form spectral analysis of Section 3.1 possible. In 2D, the derivative $\partial\sigma^*/\partial\epsilon^*$ evolves with the strain state and $\boldsymbol{AC}$ can no longer be characterized analytically. The following section examines whether the tangent-consistent plus arc-length combination retains its necessity properties in this setting.

3.3. 2D case - Bar with a notch

The analysis presented earlier is now performed on a 2D example. A bar with a notch is considered - the same example will be revisited in Section 6.1 to demonstrate the regularization. The geometry and the loading used are presented in Figure 9. The left end of the bar is fixed, top end of the bar is constrained to move horizontally, and a horizontal displacement is applied at the right end. Since there is no length scale yet, the damage is expected to localize over a one-element-wide region. The severely brittle snapback this geometry produces is deliberate: the 1D analysis of Section 3.1 shows that the choice of hyper-parameter and continuation strategy



(a) Force vs Displacement curves for two mesh sizes.



(b) Strain distribution along the bar corresponding to the final load step in the force-displacement response in Figure 8a.

Figure 8: Results for the 1D bar with a body force without regularization.

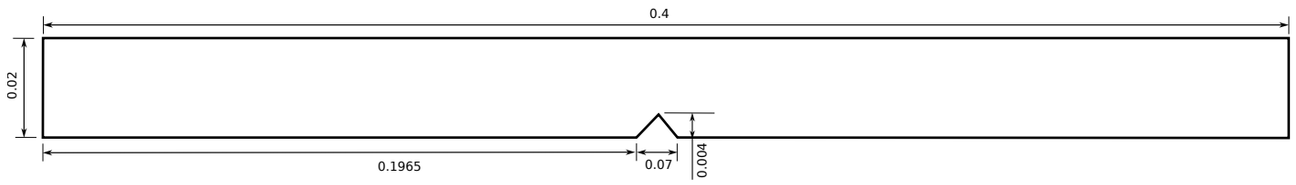


Figure 9: Geometry of the bar with a notch. All dimensions are in meters.

matters most precisely in the snapback regime, so a sufficiently brittle 2D example is required to expose the differences between the three scenarios defined below.

Definition 2 (Scenarios)**.** *The scenarios tested in section 3.1 are considered in 2D, but now without the branch restrictions, as follows:*

1. *Degraded elastic moduli of the corresponding material states are taken as the hyper-parameters. The system is displacement driven and an arc-length solver has not been used. This corresponds to cases 1 and 3 tested in section 3.1, with* $\boldsymbol{C}_2 = g(D)\boldsymbol{E}$.
2. *Tangent moduli of the corresponding material states are taken as the hyper-parameters. The system is displacement driven and an arc-length solver has not been used. This corresponds to cases 1 and 3 tested in section 3.1, with* $\boldsymbol{C}_2$ *equal to the tangent supplied from the material update.*
3. *Tangent moduli of the corresponding material states are taken as the hyper-parameters. Arc-length solver has been used. This corresponds to case 2 tested in section 3.1, with* $\boldsymbol{C}_2$ *equal to the tangent supplied from the material update.*

3.3.1. Material update

The material updates take the irreversibility into consideration using a damage variable D^n and a history variable $\mathcal{H}^n = \max_{k\leq n} \psi_0(\boldsymbol{\epsilon}^k)$, defined using the converged solution at load step n. Here, $\psi_0(\boldsymbol{\epsilon}) = \frac{1}{2}\boldsymbol{\epsilon}^T\boldsymbol{C}\boldsymbol{\epsilon}$, $\boldsymbol{C}$ is the elastic stiffness tensor. The update is performed at each Gauss point during an iteration i of load step $n+1$ using D^n and $\mathcal{H}^n$.

The constitutive model, detailed in Appendix B, is the linear-softening damage law $\boldsymbol{\sigma} = g(D)\boldsymbol{C}\boldsymbol{\epsilon}$, with degradation function g defined in equation 23. The damage evolution is

$$D = \frac{\sqrt{k\psi_0} - \sqrt{Y_c}}{(k-1)\sqrt{Y_c}}, \tag{25}$$

where k, Y_c are material parameters. Damage initiates when $\psi_0 \geq Y_c/k$. The 1D relation in equation 24 can be recovered by taking $Y_c = 1/2E\epsilon_i\epsilon_f$ and $k = \epsilon_f/\epsilon_i$.

At iteration i of load step $n+1$, given the current mechanical state $(\boldsymbol{\epsilon}, \boldsymbol{\sigma})$ at a Gauss point, the material update proceeds as follows:

1. If $\psi_0(\boldsymbol{\epsilon}) < \mathcal{H}^n$: the element is unloading. The material state is the closest point on the elastic branch $\boldsymbol{\sigma}^* = g(D^n)\boldsymbol{C}\boldsymbol{\epsilon}^*$.
2. If $\mathcal{H}^n \leq \psi_0(\boldsymbol{\epsilon}) < Y_c/k$: the element is reloading toward the yield point, no new damage. Same as case 1.
3. If $\psi_0(\boldsymbol{\epsilon}) \geq \max(Y_c/k, \mathcal{H}^n)$ and $D^n < 1$: damage may increase. In this case, the steps followed during the material update are given in algorithm 2 in Appendix A.

   - The closest point is computed on the elastic (degraded) branch $\boldsymbol{\sigma}^* = g(D^n)\boldsymbol{C}\boldsymbol{\epsilon}^*$ as outlined in Part-A in Algorithm 2.

- The closest point is computed on the softening branch $\boldsymbol{\sigma}^* = g(D(\boldsymbol{\epsilon}^*))\boldsymbol{C}\boldsymbol{\epsilon}^*$ as outlined in Part-B in Algorithm 2.
- If the damage computed above exceeds the maximum damage (typically $D_{\max} = 1 - \texttt{TOL}$), the Part-C then computes the closest point on the branch $\boldsymbol{\sigma}^* = g(D_{\max})\boldsymbol{C}\boldsymbol{\epsilon}^*$.
- Part D then checks the admissibility of the computed elastic and damage material states. The solution computed in Part-A must be such that the corresponding $\psi_0 \leq \mathcal{H}^n$ and the solution from Part-B must be such that the associated damage must be larger than or equal to D^n. Then depending on which states are admissible, they then return the appropriate material state $(\boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*)$, together with the updated tangent. The tangent in this case is a second-order tensor computed using equation 53. In 1D, it reduces to $g(D)E$ on the elastic branch and to m on the softening branch.

At convergence of the AM iterations for load step $n+1$, the state variables are updated: $\mathcal{H}^{n+1} = \max(\mathcal{H}^n, \psi_0(\boldsymbol{\epsilon}^{n+1}))$ and D^{n+1} is obtained by substituting $\psi_0 = \mathcal{H}^{n+1}$ into equation 25.

Mechanical update

The 2D mechanical update is described in Section 5. The element-wise hyper-parameter supplied by the material update is used in the Euler-Lagrange equations to carry out the mechanical update.

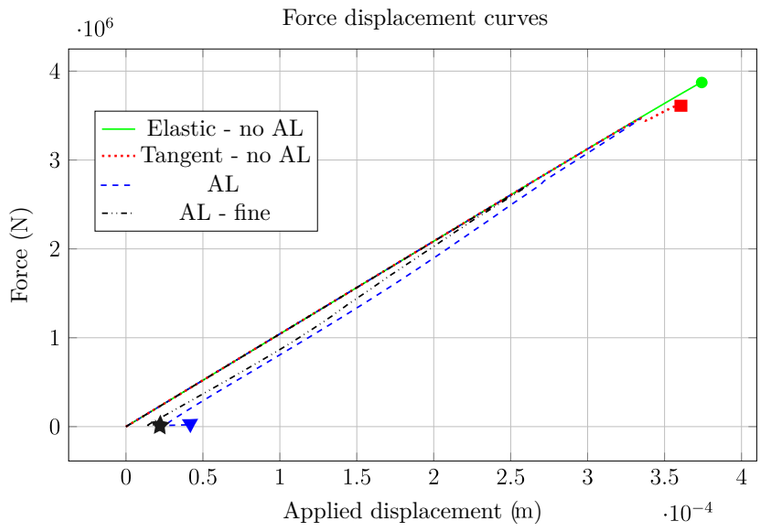


Figure 10: Load - displacement response for the bar with a notch. AL stands for Arc Length. Elastic - no AL corresponds to Figure 11a - scenario 1, Tangent - no AL corresponds to Figure 11b - scenario 2, AL corresponds to Figure 11c - scenario 3, AL - fine corresponds to Figure 11d - scenario 3. The symbols represent the final load-step at which a converged solution is obtained.

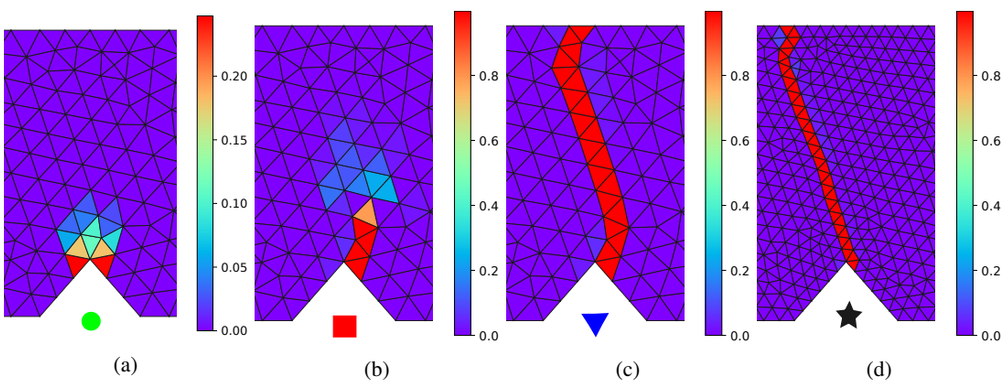


Figure 11: Damage distributions without regularization, each shown at its final converged load step. The symbol at the bottom of each panel (circle, square, star, triangle) matches the final converged load step marked on the corresponding load-displacement curve in Figure 10. (a) Coarse mesh - Damaged elastic modulus used for hyper-parameter and no AL (scenario 1), (b) Coarse mesh - Tangent modulus used for hyper-parameter and no AL (scenario 2), (c) Coarse mesh - Tangent modulus used for hyper-parameter with AL solver (scenario 3), (d) Fine mesh - Tangent modulus used for hyper-parameter with AL solver (scenario 3).

The results obtained for the damage profile at the final converged load step at the center section of the bar in Figure 9 for the three scenarios examined are presented in Figure 11 and the load-displacement response in Figure 10. In scenario 1, the damage profile is diffuse as presented in Figure 11a. This is in line with the earlier discussion regarding the path taken by the mechanical and the material states in 1D, presented in Figures 4a and 7a, where the direction of strain increment in an element may result in a transition towards a homogeneous solution branch. The corresponding load-displacement curve in Figure 10 is still almost straight when the

simulation is terminated. As in Case 1 of Section 3.1, the iterates barely move between AM steps while the distance functional remains bounded away from zero; the practical implications for discrete-database implementations are discussed in Section 3.1.

When the tangent moduli are taken as the hyper-parameter (scenario 2), the damage profile initially localizes until a certain load step, as presented in Figure 11b. After this, a snapback response is expected. Since the system is displacement driven, it is expected that the system jumps through and the fully damaged band develops over the entire section instantaneously in a single load step. However, as presented in Figure 11b, after the initial localized profile, the damage becomes diffuse as in the earlier case, and also as presented in Figure 7b. The corresponding load-displacement curve in Figure 10 reveals a similar behavior. At the point where a snapback is expected, there is a small drop in the load, which later on recovers and starts to increase again.

The next combination is the arc-length solver with the tangent moduli as the hyper-parameter (scenario 3). As in the 1D case, the damage localizes to a one-element-wide region as shown in Figure 11c. The arc length solver is able to track the severe snapback response of the bar as presented in Figure 10. This also demonstrates the robustness of the arc-length constraint, which will be described in detail later.

When similar combinations are tested for a fine mesh, the scenario 1 combination needs an extremely large number of iterations to converge, once the damage begins to increase. This is clear from the analysis in the previous section, where the spectral radius of the matrix $\boldsymbol{AC}$ gets much closer to 1 as the mesh is refined. Scenario 2 at the refined mesh (not shown) exhibits the same pre-snapback localization and the same loss of convergence at snapback. The scenario 3 case at the refined mesh, presented in Figure 11d, presents a damage band that is one-element wide, exhibiting the expected mesh sensitivity. The load-displacement curve, presented in Figure 10, shows a snapback response that is different from the coarse mesh case.

Remark. *Results from the analysis performed in appendix C reveal that an optimal rate of convergence is obtained when the metric in the distance functional is taken equal to the tangent, $\partial\sigma^*/\partial\epsilon^*$, obtained from the corresponding material state. For materials that undergo damage, this derivative is negative. Thus, during the alternating minimization process, for the elements that undergo damage, $\boldsymbol{C}$ is smaller than zero. The convergence of the solver is still judged based on the metric that equals the elastic modulus, while the tangent is used only during the alternating minimization and the corresponding Newton iterations (equations 6 and 9).*

In a non-DD setting, the problem involves minimizing a functional that is non-convex. The energy density depends on displacement and damage, and is separately convex with respect to each variable. Consequently, the minimization is carried out by alternating between the two variables. This procedure allows the minimizers to jump through instabilities such as snapbacks under displacement control (although this may require a larger number of iterations). It should be noted that using a monolithic (Newton) solver for such a non-convex problem, without additional refinements or an arc-length constraint, may lead to non-convergence [15, 17, 33, 4]. In the DD setting, however, the database resembles that in Figure 1, and thus the non-convexity is inherently embedded in the stored data. For this reason, an arc-length method has been employed to trace the entire bifurcation branch without requiring any jumps. As presented, not using an arc-length solver may result in a widened damage profile even in the absence of regularization. This may result in a mesh-dependent behavior even after the introduction of a length scale.

To conclude, with the ingredients presented (the arc length solver and the choice of hyper-parameters), the data driven solver is able to exhibit the expected mesh dependence of the solution as expected when there is no regularization. To emphasize, the emergence of localized or diffuse damage in this setting is not enforced but results entirely from the interaction between the DD hyper-parameters, continuation strategy, and material non-convexity.

4. Introduction of a length scale

In the previous section, it was shown that, in the absence of regularization, the DD algorithm exhibits mesh dependence of the solution, as expected, provided that the appropriate algorithmic ingredients are employed. The natural next step is therefore to introduce a regularization mechanism such that the results, most notably the width of the localization band, become independent of the mesh size.

In non-DD settings, this objective is commonly achieved through a variety of approaches, including gradient-enhanced damage models, nonlocal damage evolution laws, and nonlocal or gradient-based strain formulations. In the DD setting considered here, however, the direct use of damage gradients is non-trivial or even infeasible, since the damage variable, when present, is attached to the material state rather than defined as an independent field. For this reason, regularization is instead introduced through the gradient

of the mechanical strain. Furthermore, the regularization is modeled independently of the material database - here represented by the local constitutive model - in contrast to the approach adopted in [35].

To this end, the strain-based Lipschitz regularization introduced in [18] is first adapted to the data-driven framework in a one-dimensional setting. The method is then applied to the localization problem introduced in the previous section. Finally, a relaxed version of the regularization strategy is developed and presented for the two-dimensional case.

4.1. The method

The regularization technique was introduced in [18], where it was tested with a constitutive model. Here, it will first be extended for use with the DDCM methodology. To begin with, a length scale is introduced into the problem by restricting the gradient of the strain variable to lie in a certain interval. More precisely, it is required that $\epsilon_{,x} \in [-1/\ell_c, 1/\ell_c]$, where ℓ_c is a length parameter. It is expected that the introduction of this constraint prevents the spurious strain localization and the dependence of the solution on the Finite Element mesh size.

The introduction of the Lipschitz constraint on the strain variable modifies the equation of equilibrium in 1D to

$$\sigma_{,x} - \lambda_{,xx} = 0, \tag{26}$$

see equation 37 of [18], where λ is the Lagrange multiplier that is dual to the Lipschitz constraint. $\lambda \neq 0$ only in the regions of the body where the constraint is active. The equilibrium equations used in DDCM should thus take this into account.

Assuming piecewise affine interpolation for the displacement variable and thus, a piecewise constant variation of the stresses and the strains, the distance functional becomes

$$d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*) = \frac{1}{2}(\boldsymbol{B}\boldsymbol{u} - \boldsymbol{\epsilon}^*)^T \boldsymbol{C}\boldsymbol{w}(\boldsymbol{B}\boldsymbol{u} - \boldsymbol{\epsilon}^*) + \frac{1}{2}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^*)^T \boldsymbol{C}^{-1}\boldsymbol{w}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^*). \tag{27}$$

The objective is now to find the minimizers of the above functional so that the strain field satisfies the Lipschitz constraint, and the stress fields satisfy the equilibrium. The Lagrange multipliers that enforce the constraint on the strain field act as couple stresses. Since the displacement field is piecewise affine, the strain field is piecewise constant. To compute the gradient of strain, a second mesh, referred to as the lip-mesh, following the terminology of [9], is constructed using the centroids of the displacement mesh as vertices. See Figure 12, where the lip-mesh is indicated in red; the 2D construction is described in Section 5. On the lip mesh, the strain field is interpolated in a piecewise affine and continuous manner, giving a piecewise constant strain gradient over each lip-mesh element. The solution is now

$$\text{Sol} \in \arg \min_{(\boldsymbol{\epsilon},\boldsymbol{\sigma})\in\mathcal{E},(\boldsymbol{\epsilon}^*,\boldsymbol{\sigma}^*)\in D} d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*), \tag{28}$$

where the equilibrium manifold is now defined as (in discrete form)

$$\mathcal{E} = \{(\boldsymbol{\epsilon}, \boldsymbol{\sigma}) : \boldsymbol{\epsilon} = \boldsymbol{B}\boldsymbol{u}, \boldsymbol{\epsilon} \text{ is such that } |\boldsymbol{B}'\boldsymbol{\epsilon}| \leq 1/\ell_c, \boldsymbol{u} = \boldsymbol{u}_D \text{ on } \partial\Omega_d, \text{ and } \boldsymbol{B}^T\boldsymbol{w}\boldsymbol{\sigma} + \boldsymbol{P}^T\boldsymbol{\lambda} = 0\}, \tag{29}$$

with $\boldsymbol{\lambda}$ the vector of Lagrange multipliers, $\boldsymbol{B}'$ the discrete gradient operator on the strain variable (see Figure 12), and $\boldsymbol{P} = \boldsymbol{B}'\boldsymbol{B}$ the discrete second gradient operator. The term $\boldsymbol{P}^T\boldsymbol{\lambda}$ carries an implicit element-length weighting analogous to $\boldsymbol{w}$ on the first term; this weighting is absorbed into $\boldsymbol{\lambda}$ in the 1D setting and written explicitly in the 2D formulation. Given the peak stress value, and thus the value of Y_c/k in equation 25, ℓ_c can be related to the amount of energy dissipated in the (1D) body from the initiation of damage until complete failure. In [18, eq. 55], the gradient limiter ℓ_c is related to the fracture energy G_c as $G_c = Y_c\ell_c$.

The constraints $|\boldsymbol{P}\boldsymbol{u}| \leq 1/\ell_c$ are interpreted element wise. That is

$$|\boldsymbol{P}\boldsymbol{u}| \leq 1/\ell_c \implies |P_{ij}u_j| \leq 1/\ell_c \quad \forall \quad i = 1, 2, 3, \ldots N, \tag{30}$$

where N denotes the number of lip elements (see Figure 12). By splitting the constraints $|\boldsymbol{P}\boldsymbol{u}| \leq 1/\ell_c$ as $\boldsymbol{P}\boldsymbol{u} \leq 1/\ell_c$ (plus side) and $-\boldsymbol{P}\boldsymbol{u} \leq 1/\ell_c$ (minus side), the Lagrangian can now be defined as:

$$\Pi = d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*) - \boldsymbol{\eta}^T \boldsymbol{B}^T \boldsymbol{w}\boldsymbol{\sigma} + \sum_{i=1}^{N_c^+} \lambda_i^{u+} \left((\boldsymbol{P}\boldsymbol{u})_i - \frac{1}{\ell_c} \right) + \sum_{i=1}^{N_c^+} \lambda_i^{\eta u+} \left((\boldsymbol{P}(\boldsymbol{u}+\boldsymbol{\eta}))_i - \frac{1}{\ell_c} \right) + \sum_{i=1}^{N_c^-} \lambda_i^{u-} \left(-(\boldsymbol{P}\boldsymbol{u}) - \frac{1}{\ell_c} \right) + \sum_{i=1}^{N_c^-} \lambda_i^{\eta u-} \left(-(\boldsymbol{P}(\boldsymbol{u}+\boldsymbol{\eta}))_i - \frac{1}{\ell_c} \right). \quad (31)$$

Here, N_c^+ and N_c^- denote the number of active plus and minus constraints, respectively, and λ_i^u, $\lambda_i^{\eta u}$ are the Lagrange multipliers dual to the constraint on the displacements $\boldsymbol{u}$ and the perturbed displacements $\boldsymbol{u}+\boldsymbol{\eta}$, respectively. $(\bullet)_i$ denotes the ith component of the vector $\bullet$. $\boldsymbol{\lambda}^u$, $\boldsymbol{\lambda}^{\eta u}$ are thus defined as piecewise constant on the lip elements.

The solution is defined as the collection that renders the Lagrangian stationary.

$$\Pi \leftarrow \text{stat!} \quad (32)$$

The third and the fourth terms in the Lagrangian warrant some discussion. It is clear that the third term, together with the condition that $\lambda^u \geq 0$, enforces the inequality constraint $|\epsilon_{,x}| \leq 1/\ell_c$. Treating $\boldsymbol{\eta}$ as the virtual displacement, the fourth term enforces the constraint that in the regions where the inequality constraint is satisfied as an equality, the perturbed field, $\boldsymbol{u}+\boldsymbol{\eta}$, must also satisfy the constraint as an equality. A priori, there is no restriction on the sign of $\lambda^{\eta u}$. However, as demonstrated below, $\lambda_i^{\eta u} = -\lambda_i^u$ at convergence. Since λ^u is already restricted, this relationship inherently determines the sign of $\lambda^{\eta u}$.

For notational convenience, the minus constraints will be dropped, since the only difference they make is in the sign of the couple stress. The Euler Lagrange equations can be seen to be

$$\boldsymbol{r}_{\boldsymbol{u}} = \boldsymbol{B}^T \boldsymbol{C}\boldsymbol{w}\boldsymbol{B}\boldsymbol{u} - \boldsymbol{B}^T \boldsymbol{C}\boldsymbol{w}\boldsymbol{\epsilon}^* + \boldsymbol{P}^T \boldsymbol{\lambda}^u + \boldsymbol{P}^T \boldsymbol{\lambda}^{\eta u} = 0, \quad (33)$$

$$\boldsymbol{r}_{\boldsymbol{\eta}} = -\boldsymbol{B}^T \boldsymbol{w}\boldsymbol{C}\boldsymbol{B}\boldsymbol{\eta} - \boldsymbol{B}^T \boldsymbol{w}\boldsymbol{\sigma}^* + \boldsymbol{P}^T \boldsymbol{\lambda}^{\eta u} = 0, \quad (34)$$

$$r_{\lambda_i^u} = (\boldsymbol{P}\boldsymbol{u})_i - \frac{1}{\ell_c} = 0, \quad (35)$$

$$r_{\lambda_i^{\eta u}} = (\boldsymbol{P}(\boldsymbol{u}+\boldsymbol{\eta}))_i - \frac{1}{\ell_c} = 0. \quad (36)$$

At convergence, $\boldsymbol{\epsilon} = \boldsymbol{\epsilon}^*$ and $\boldsymbol{\eta} = 0$. Thus, from equation 33, it can be seen that $\lambda_i^u = -\lambda_i^{\eta u}$, as mentioned earlier. The equations of equilibrium can be recovered as $\boldsymbol{B}^T \boldsymbol{w}\boldsymbol{\sigma} + \boldsymbol{P}^T \boldsymbol{\lambda}^u = 0$, since $\boldsymbol{\sigma} = \boldsymbol{\sigma}^*$ at convergence.

The strain variable is interpolated on the lip-mesh with the same polynomial order as the displacement variable on the displacement mesh (piecewise affine and continuous in both cases). The stability of this combination has been demonstrated in Appendix F.2. A second, separate stability criterion arises from the active-set multipliers λ^u that enforce the inequality constraint on the strain gradient. The regularization restricts the strain gradient to a piecewise-constant field on the lip-mesh, so the associated Lagrange multipliers are taken to be piecewise constant over the lip-mesh. The same lip-mesh construction and interpolation pairing has been employed in the Lipfield formulation [9]. A formal inf-sup analysis of this active-set formulation is beyond the scope of the present work; instead, the stability of the discrete solution is assessed by refining the mesh and observing the variations, if any, in the results - a numerical proxy for well-posedness that is presented for both the 1D and 2D examples in the following sections.

4.2. Application and results

The regularization technique is now applied to the 1 m long 1D bar tested earlier with a body force. The value of ℓ_c is taken to be 0.1m. Arc length solver has been used like in the previous case with similar loading conditions. A piecewise linear and continuous interpolation has been used for displacement and a piecewise constant for the mechanical stress. A lip-mesh had been used to impose the constraint on the strain gradient. The strain is taken to be piecewise affine and continuous on the lip-mesh. See Figure 12.

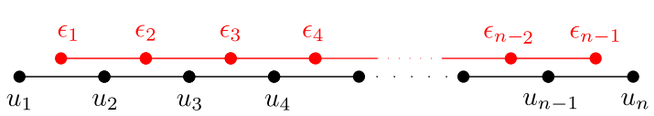
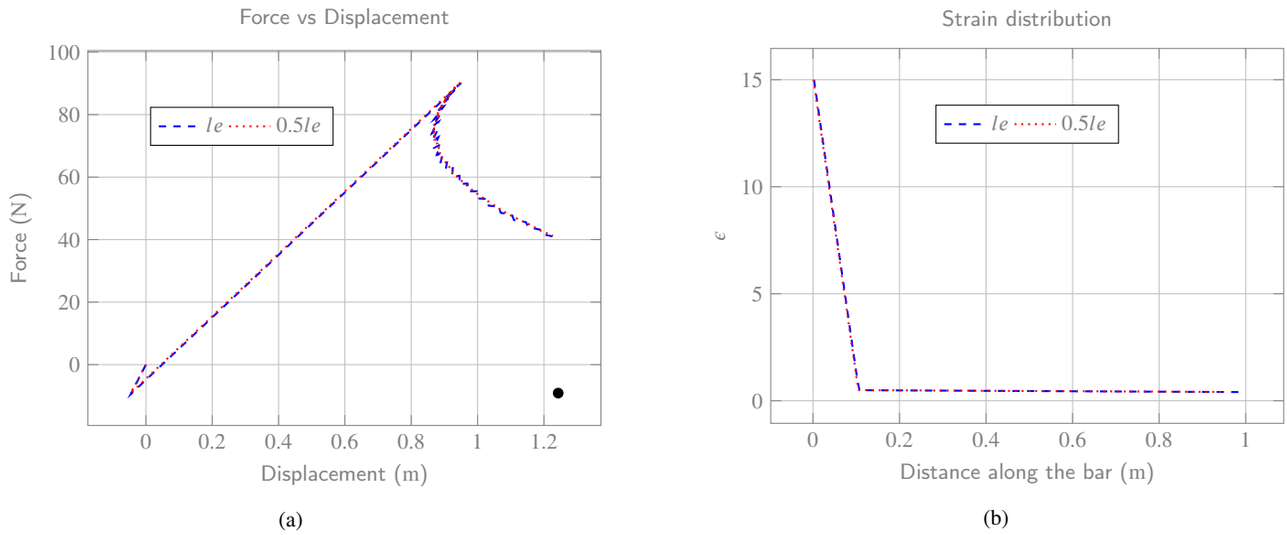


Figure 12: Mesh for displacement and strain

The load displacement curves and the strain distribution at failure have been presented in Figures 13a and 13b for two different mesh sizes (compare with Figure 8). Several observations warrant further discussion. First, as discussed in [18], the constraint must



(a) (b)

Figure 13: Results with regularization. (a) Force vs Displacement curves for two mesh sizes. (b) Strain distribution along the bar.

be removed from the elements that are fully damaged to allow for the strains to localize in the fully damaged element and prevent the expansion of damage further into the body. The black dot in Figure 13a corresponds to the case when the constraint is deactivated; the force-displacement curve then jumps to zero. This deactivation is an artificial device: physically, a fully damaged element should transmit no stress, but abruptly removing the regularization creates a discontinuity in the model. In 1D, this discontinuity is manageable because failure is contained in a single element and the deactivation affects only that element's constraint. See [16, 31, 23] and the references therein for similar discussions where the damaged regions extend across the body and various modifications/ adaptations of the corresponding regularization method have been proposed to force a localized damage profile. Second, taking a closer look at the force-displacement curve as in figure 14, a number of smaller snapbacks can be observed. It can be explained as follows: since the regularization is introduced through an inequality constraint, and this constraint is numerically resolved with a finite mesh size, the regularization is inactive until strain localizes in the some region of the body. The strains then grow locally till the constraint becomes active. Once activated, the constraint expands the damage zone further into the neighboring elements, where the strains again grow locally. This process carries on until some part of the body is fully damaged. Thus, the problem switches between a local (albeit with a varying length scale) and non-local problem as the constraint is activated and the damage spreads into other elements. The snapback occurs when the strains localize and the recovery corresponds to the activation of the constraint. It can also be seen that the amplitude of this sawtooth like pattern decreases as the mesh is refined, but the number of tooth increases.

Remark. *In hindsight, deactivating the constraint only after an element is fully damaged does indeed cause convergence issues in 2D. This issue has been discussed in multiple references - in the context of non-local damage models in [16]. Thus, for the 2D case, the constraint is gradually weakened as the damage value increases. Also, to reduce the impact of sawtooth like pattern, the inequality constraint is weakened and a log-barrier-like function is used instead. More details can be found in the sections below.*

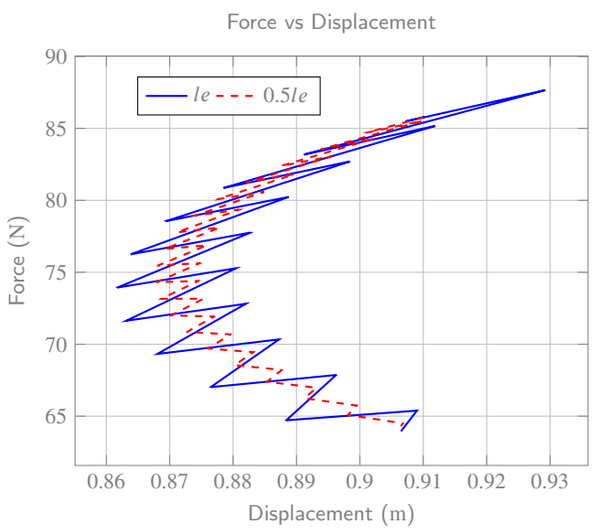


Figure 14: Force vs Displacement curves for two mesh sizes zoomed in.

Thus, in the 1D case, the proposed regularization technique ensures that the result remains independent of the mesh size, as expected. Though not reported here, the simulations performed in 2D case reveal convergence issues if the constraint is abruptly deactivated once an element fails completely ($d = 1$). Also, the activation-deactivation of elements lead to sawtooth like pattern in the load displacement curve, which may prove tedious to trace in 2D. To address these challenges, an alternative approach is presented below for the 2D case, in which the regularization is gradually weakened as the damage increases.

5. 2D Case

As mentioned in the previous section, the strain based Lipschitz constraint introduces a length scale into the problem and results in a mesh independent solution. However, as was shown, the inequality constraint introduces a saw-tooth like pattern in the force-displacement response, which can be tedious to track. Also, to ensure that the damage zone does not extend across the body, the constraint (and thus, the length scale) will have to be removed from the fully damaged regions of the body. Abrupt removal of the constraint will likely lead to issues with convergence as well. Thus, two modifications will be introduced in this section in its extension to 2D. First, the constraint will be gradually weakened as the damage level increases. Second, the inequality constraint will be replaced by a log-barrier-like function. It is expected that the log-barrier like regularization will result in a smaller number of peaks in the saw-tooth pattern reported in the 1D case, if not completely eliminate it. It is also expected that the log-barrier like form delays the effect of the regularization term until the strains start to localize and not become active right from the earlier stages before strain localization occurs, see Figure 23a.

The body is assumed to be triangulated by constant strain triangles, referred to as the primary mesh. Consequently, the displacement is piecewise linear and continuous across the elements, while the stress field is piecewise constant in each element. Thus, each element is associated with a single mechanical and material state. To impose the constraints on the gradient of strain, a lip mesh is introduced. The lip mesh consists of another set of piecewise affine triangles defined using the centroids of the triangles of the primary mesh as vertices. The strain field is then interpolated over the lip mesh in a piecewise linear manner. The gradient of strain, is thus piecewise constant over the triangles of the lip-mesh.

The distance functional remains as

$$d(\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*) = \frac{1}{2}(\boldsymbol{Bu} - \boldsymbol{\epsilon}^*)^T \boldsymbol{Cw}(\boldsymbol{Bu} - \boldsymbol{\epsilon}^*) + \frac{1}{2}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^*)^T \boldsymbol{C}^{-1}\boldsymbol{w}(\boldsymbol{\sigma} - \boldsymbol{\sigma}^*). \tag{37}$$

Similar to the 1D case, the solution is then defined as the one that renders the following Lagrangian stationary.

$$\Pi = d - \boldsymbol{\eta}^T \boldsymbol{B}^T \boldsymbol{w}\boldsymbol{\sigma} - \mu \sum_{i=1}^{N_c} w_i' \ln s_{ui} + \mu \sum_{i=1}^{N_c} w_i' \ln s_{\eta i} + \sum_{i=1}^{N_c} \lambda_i^u \left\{ s_{ui} - \left[\frac{1}{\ell_c^2} - \phi_i(\boldsymbol{u}) \right] \right\} + \sum_{i=1}^{N_c} \lambda_i^{\eta u} \left\{ s_{\eta i} - \left[\frac{1}{\ell_c^2} - \phi_i(\boldsymbol{u} + \boldsymbol{\eta}) \right] \right\}. \tag{38}$$

The function ϕ is the strain-gradient term, as defined in equation 2.3 of [25]. In the current case, it is defined as

$$\phi = 3\, k_{11i} k_{11i} + 3\, k_{22i} k_{22i} + \frac{1}{2}\, k_{12i} k_{12i}, \tag{39}$$

where $k_{lmn} = \epsilon_{lm,n}$ and Einstein summation convention has been used. Slack variables $s \geq 0$ have been introduced. Using Lagrange multipliers λ^u, the slack variables are defined as the difference between the length term and the strain gradient term. The log term stays small as the slack variables are large and begins to take over as the constraint is approached. See appendix F. This introduces the length scale into the problem. Inspired from the 1D case as in equation (31), a term involving the 'virtual displacements' $\boldsymbol{\eta}$ is also included in the above equation in the strain gradient term ϕ in addition to the one involving displacement. By taking variations with respect to $\boldsymbol{\eta}$, the equilibrium equation can be obtained, which consists of the Cauchy stress $\boldsymbol{\sigma}$ and the Lagrange multipliers $\lambda^{\eta u}$ that act as couple stresses. The modified equilibrium equation because of the strain gradients is thus taken care of automatically.

The displacement field satisfies Dirichlet boundary conditions - $\boldsymbol{u} = \boldsymbol{u}_D$ on $\partial\Omega_D$. The test function $\boldsymbol{\eta}$ is equal to $\boldsymbol{0}$ on $\partial\Omega_D$. The factors w_i' denotes the area of the lip elements and is thus similar to $\boldsymbol{w}$ in equation (3). When the constraints are to be relaxed gradually, the Lagrangian becomes

$$\Pi = d - \boldsymbol{\eta}^T \boldsymbol{B}^T \boldsymbol{w}\boldsymbol{\sigma} - \mu \sum_{i=1}^{N_c} w_i' \ln s_{ui} + \mu \sum_{i=1}^{N_c} w_i' \ln s_{\eta i} + \sum_{i=1}^{N_c} \lambda_i^u \left\{ s_{ui} - \left[\frac{1}{\ell_c^2} - g(d_i)\phi_i(\boldsymbol{u}) \right] \right\}$$

$$+\sum_{i=1}^{N_c}\lambda_i^{\eta u}\left\{s_{\eta i}-\left[\frac{1}{\ell_c^2}-g(d_i)\phi_i(\boldsymbol{u}+\boldsymbol{\eta})\right]\right\}, \quad (40)$$

where g is a function such that $g(0)=1$ and $g(1)=0$. The function ϕ from equation (39) can be written in discrete form over a lip element i as

$$\phi_i(\boldsymbol{u})=\boldsymbol{u}^T\boldsymbol{G}_i^T\boldsymbol{G}_i\boldsymbol{u}, \quad (41)$$

where $\boldsymbol{G}_i$ denotes the discrete second gradient operator, written on the ith lip element.

Remark. *A couple of observations are warranted. First, the function g is selected so that* $g(0)=1$ *and* $g(1)=0$*. This means that when there is no damage, the regularization is fully effective, and when fully damaged, the regularization is removed. The transition between the two states is gradual if g is smooth. Second, in this study, the parameter* μ *is fixed to be a finite positive value. This method resembles the interior point method for resolving inequality constraints in a standard optimization setting, where successive problems are solved with decreasing* μ*,* $\mu\to 0$*. Thus, as the parameter* μ *is made smaller, the method presented here boils down to the inequality case and the problems/ features about snapback that were observed in the 1D case return. It is expected that taking* μ *to be finite and fixed prevents this issue. A systematic study of the influence of* μ *is left to future work, as the present focus is on establishing that a smoothed regularization can be integrated with the DDCM algorithm.*

The displacement vector $\boldsymbol{u}$ is split into degrees of freedom that are unknown or free, $\boldsymbol{u}_f$, and the part that follows from the applied boundary conditions, $\boldsymbol{u}_g$. Specifically, if $\boldsymbol{u}\in\mathbb{R}^N$, $\boldsymbol{u}_g\in\mathbb{R}^N$ is a vector of zeros and ones, with ones at places corresponding to the nodes where the boundary conditions are applied. The vector $\boldsymbol{u}_f\in\mathbb{R}^{N_f}$ will be determined from the solution of the problem. Thus, $\boldsymbol{u}=\boldsymbol{C}_f\boldsymbol{u}_f+\lambda_g\boldsymbol{u}_g$, where $\boldsymbol{C}_f\in\mathbb{R}^{N\times N_f}$ is a matrix that relates the total and the free degrees of freedom, $\lambda_g\in\mathbb{R}$ is the magnitude of the applied boundary condition on $\partial\Omega_D$. Since $\boldsymbol{\eta}=\boldsymbol{0}$ on $\partial\Omega_D$, it is simply expressed as $\boldsymbol{\eta}=\boldsymbol{C}_f\boldsymbol{\eta}_f$. Using the expressions for $\boldsymbol{u}$ and $\boldsymbol{\eta}$ in the Lagrangian and taking its variations results in the Euler Lagrange equations.

$$\boldsymbol{r}_{\boldsymbol{u}_f}=\boldsymbol{C}_f\boldsymbol{B}^T\mathcal{C}\boldsymbol{w}(\boldsymbol{B}\boldsymbol{u}-\boldsymbol{\epsilon}^*)+\sum_{i=1}^{N_c}2\lambda_i^u g(d_i)\boldsymbol{C}_f\boldsymbol{G}_i^T\boldsymbol{G}_i\boldsymbol{u}+\sum_{i=1}^{N_c}2\lambda_i^{\eta u}g(d_i)\boldsymbol{C}_f\boldsymbol{G}_i^T\boldsymbol{G}_i(\boldsymbol{u}+\boldsymbol{\eta})=\boldsymbol{0}, \quad (42)$$

$$\boldsymbol{r}_{\boldsymbol{\eta}_f}=-\boldsymbol{C}_f\boldsymbol{B}^T\boldsymbol{w}(\boldsymbol{\sigma}^*+\mathcal{C}\boldsymbol{B}\boldsymbol{\eta})+\sum_{i=1}^{N_c}2\lambda_i^{\eta u}g(d_i)\boldsymbol{C}_f\boldsymbol{G}_i^T\boldsymbol{G}_i(\boldsymbol{u}+\boldsymbol{\eta})=\boldsymbol{0}, \quad (43)$$

$$r_{\lambda_u^i}=\boldsymbol{u}^T\boldsymbol{G}_i^T\boldsymbol{G}_i\boldsymbol{u}-\frac{1}{\ell_c^2}+s_{ui}=0, \quad (44)$$

$$r_{\lambda_\eta^i}=(\boldsymbol{u}+\boldsymbol{\eta})^T\boldsymbol{G}_i^T\boldsymbol{G}_i(\boldsymbol{u}+\boldsymbol{\eta})-\frac{1}{\ell_c^2}+s_{\eta i}=0, \quad (45)$$

$$r_{s_{ui}}=\lambda_i^u s_{ui}-\mu w_i'=0, \quad (46)$$

$$r_{s_{\eta i}}=\lambda_i^{\eta u}s_{\eta i}-\mu w_i'=0. \quad (47)$$

The above system is resolved for $\boldsymbol{u}_f,\boldsymbol{\eta}_f,\lambda^u,\lambda^{\eta u},s_u,s_\eta$, knowing λ_g. When arc length solver is used, λ_g is treated as an unknown and the system is supplemented by the arc-length equation. The tangent matrix used during the Newton iterations are given in Appendix B. The details of the arc-length solver can be found in Appendix E.

Remark. *As in the 1D case, at convergence,* $\boldsymbol{\epsilon}=\boldsymbol{\epsilon}^*$*, and* $\boldsymbol{\sigma}=\boldsymbol{\sigma}^*$*. Thus,* $\lambda_i^{\eta u}=-\lambda_i^u$*. Also, the Lagrange multipliers that take the role of couple stresses are computed as* $\lambda_i^u=\mu w_i'/s_{ui}>0$ *since* $s_{ui}>0$ *as a consequence of the log-barrier-like regularization.*

6. Applications

The formulation presented above will be applied for the case of a bar with a notch and an L-beam. The intent is to examine the effectiveness of the regularization technique in terms of making the solution mesh independent.

6.1. A bar with a notch

The regularization technique described above has been applied to a notched bar problem, as illustrated in Figure 9. The left end of the bar is fixed, top end of the bar is constrained to move horizontally, and a horizontal displacement has been applied at the right end. The displacement magnitude is controlled using the arc-length method, which is detailed in the Appendix E. The displacement field is discretized using constant-strain triangular elements, as described earlier. The strain variable is subsequently interpolated onto a lip

mesh, which is then used to evaluate the strain gradient. An example of the meshes used for the displacement and the strain variables is presented in Figure 23b.

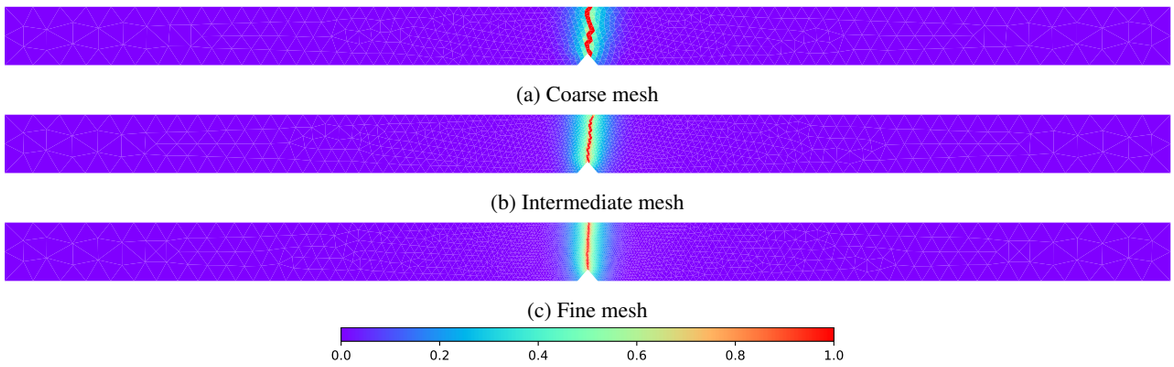


Figure 15: Damage distribution for the notched bar at the final load step, for three mesh sizes

The geometry of the bar is shown in Figure 9. The material parameters used for this study were $E = 210\,\text{GPa}, \nu = 0.3, Y_c = 2 \times 10^6\,\text{J/m}^3, k = 10, \ell_c = 0.5\,\text{m}$, see Appendix B. Because the strain-gradient term ϕ is modulated by the damage-dependent function g, the regularization strength dynamically evolves during the simulation. Consequently, the effective regularization length is significantly smaller than the prescribed parameter ℓ_c, as demonstrated in the following examples. The parameter $\mu = 2 \times 10^6\,\text{J}$ in equation (40). The function g is taken same as in equation (23).

Figure 15 shows the damage distribution attached to the material states associated with each element. The results are for three mesh sizes, where the ratio of element sizes of the coarse mesh to the intermediate mesh, and of the intermediate mesh to the fine mesh is 2. Compared with Figure 11, the damage profile is spread over multiple elements and is mesh independent. The variation of damage value across the center of the bar, from $(0.08\,\text{m}, 0.01\,\text{m})$ to $(0.12\,\text{m}, 0.01\,\text{m})$, where the origin is at the bottom left corner of the bar, can be seen in Figure 16. Force-displacement curves for all the three mesh sizes can be seen in Figure 17. The results are different from the ones presented in Figure 10 without any regularization.

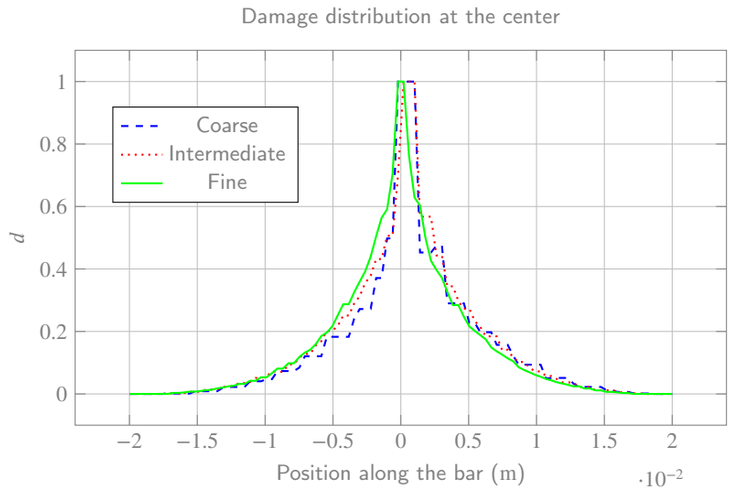


Figure 16: Variation of damage along the center of the bar, $y = 0$, $x \in [-0.02, 0.02]$ m.

It can be seen that in all the cases, the results are independent of the mesh size. The width of the damage zone does not change as the mesh is refined. Likewise, the force-displacement curve also remains the same for all the mesh sizes. A severe snap-back can be seen in the Figure 17, which has been captured by the arc length solver. The saw-tooth-like pattern observed in the 1D case (where the constraint is strictly enforced) is also present in this scenario. Perhaps testing with different values of μ in the solver or using a different arc length solver might smoothen the results. This has not been pursued further in this article and can be the focus of future work.

Damage models used in the constitutive-model setting typically employ a tension-compression split of the strain energy [1, 24] to ensure that the damage variable does not increase under compressive loading. In the DDCM setting, a tension-compression split would most naturally be embedded in the database itself, through a stress-strain response resolved separately under tensile and compressive states, rather than through a change to the distance-minimization procedure. How such a split should be reflected in the database, and

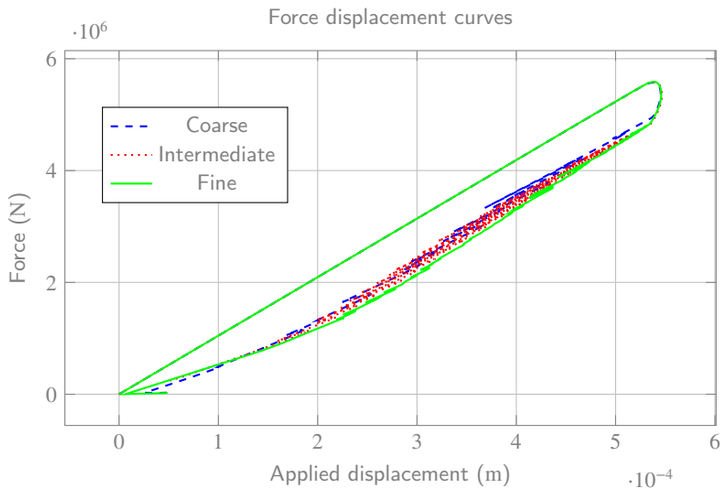


Figure 17: Force displacement curves of the notched bar for three different meshes.

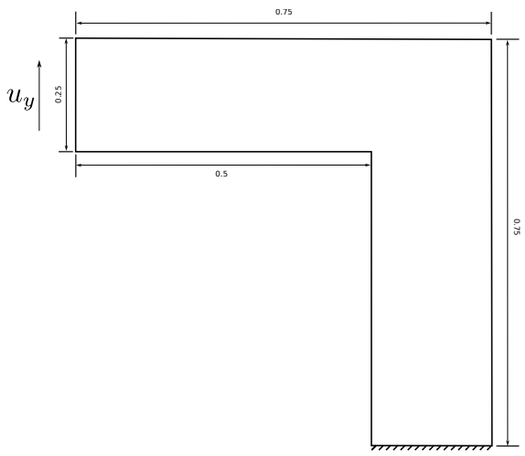


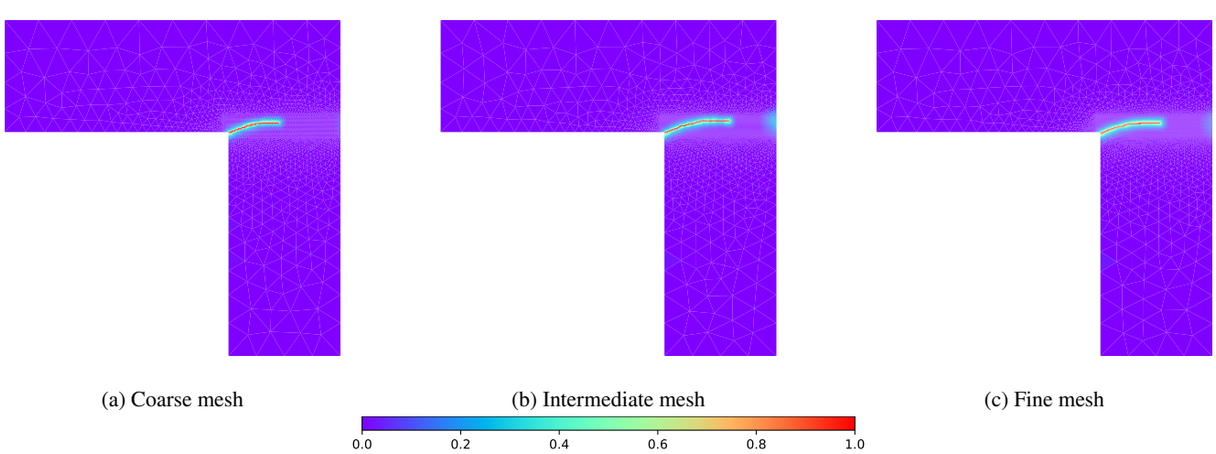
Figure 18: Geometry of the L-beam. All dimensions are in meters.

the form of the resulting data structures, raises questions that are beyond the scope of this article and are left as a direction for future work.

6.2. L Beam

The regularization technique has also been applied to an L-shaped beam (L-beam) problem. This example allows to examine the regularization effects when crack path is curved. The horizontal face of the beam is fully constrained, enforcing $u_x, u_y = 0$, while a prescribed vertical displacement was applied on the vertical face, see Figure 18. The magnitude of this displacement is controlled using the arc-length method to accurately capture the nonlinear response and post-peak behavior.

The material parameters are the same as the previous example except $\ell_c = 1$ m. As stated earlier, since the strain-gradient term ϕ is modulated by the damage-dependent function g, the regularization strength dynamically evolves during the simulation. Consequently, the effective regularization length is significantly smaller than the prescribed parameter ℓ_c. A crack is expected to initiate at the inner corner of the L-beam due to the pronounced stress concentration in this region.

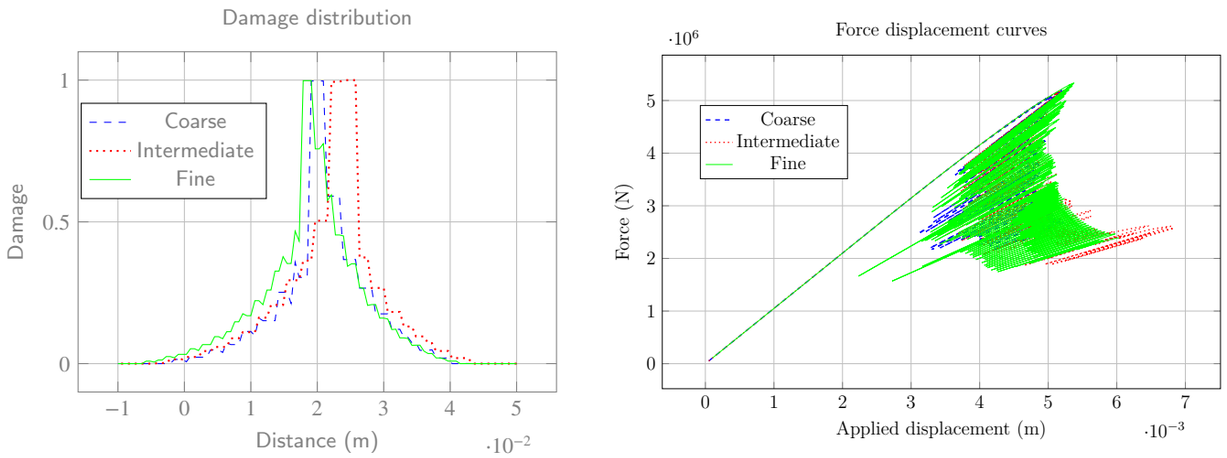


(a) Coarse mesh (b) Intermediate mesh (c) Fine mesh

Figure 19: Damage distribution for the L-beam at the final load step, for three mesh sizes

Simulations have been performed using coarse, intermediate, and fine meshes. The corresponding characteristic element sizes in the damaged region, computed as $4A/\sqrt{3}$, where A is the area of the triangle in that region, are 0.0056 m, 0.0048 m, 0.0036 m for the coarse, intermediate, and fine mesh, respectively. The resulting damage fields for the three discretizations are shown in Figure 19.

To further examine mesh sensitivity, the variation of the damage variable along a vertical section at $x = 0.08\,\text{m}$, $y \in [-0.01\,\text{m}, 0.05\,\text{m}]$ is plotted in Figure 20a. The width of the damage zone remains nearly constant across all meshes, indicating satisfactory mesh objectivity. The position of the maximum damage varies slightly, with the peaks for the intermediate and fine meshes differing by approximately 0.005 m. The force-displacement responses for all three mesh levels are compared in Figure 20b. The overall behavior is consistent across discretizations, confirming convergence of the global response. Nevertheless, the characteristic sawtooth pattern observed previously in the one-dimensional test case reappears, despite the use of the interior-point regularization to relax the strict inequality constraint. This behavior suggests that some degree of non-smoothness still remains in the progressive damage evolution.



(a) (b)

Figure 20: Results for the L beam. (a) Variation of damage along the line $x = 0.08$ m, $y \in [-0.01, 0.05]$ m. The inner corner of the L-beam is taken as the origin. (b) Force displacement curves for three different meshes.

As noted for the notched bar, the database used throughout this study is tension-compression symmetric. In the L-beam geometry, this creates a specific limitation: the loading generates a compressive zone on the right side of the beam, and the symmetric database then allows the damage variable to grow in that region. Since compression-driven damage is unphysical, the simulation was terminated at the load step where damage in the compressive zone began to increase, in contrast to the notched bar, which could be driven to complete failure.

7. Discussion

The present study analyzed the DDCM alternating minimization algorithm under damage-induced softening by employing a constitutive model as an implicit infinite database. This choice isolates algorithmic effects from database sampling artifacts and enables a closed-form spectral analysis in 1D. Since a discrete database is a sampling of the underlying model, the criteria identified here are expected to carry over to the finite-database setting, where sampling-related effects can then be examined separately.

The results demonstrate that, in problems involving material softening, the choice of hyper-parameters and continuation strategy in data-driven solvers influences not only convergence rates but also the physical character of damage localization. In particular, the combination of an arc-length solver with tangent-based hyper-parameters is shown to be essential for correctly capturing localization and snap-back behavior. This choice yields robust convergence properties that are insensitive to mesh refinement, while alternative combinations may lead to diffuse damage patterns or a complete loss of convergence as the mesh is refined. Tangent-based hyper-parameters may not always be positive definite. Thus, these set of hyper-parameters may be reserved for the mechanical update process in the corresponding Euler-Lagrange equations and a positive definite secant modulus can be used for hyper-parameters during the material update. This may lead to a loss in the rate of convergence, but preserves the instability tracking ability of the hyper-parameter - arc-length combination while dealing with discrete databases. This aspect has not been investigated and is a possible future research direction.

The need for algorithmic modifications in the presence of instabilities has been recognized previously in the data-driven literature. Kuang et al. [22], for example, considered instabilities arising from geometric nonlinearities and demonstrated that appropriate modifications to the solution strategy are required to correctly trace snap-back responses. In an analogous manner, the present study shows that employing the local material tangent as a hyper-parameter allows the data-driven equilibrium operator to develop zero eigenvalues when a snap-back response is expected. When combined with an arc-length solver, this modification enables the data-driven formulation to capture material instabilities in a manner consistent with softening behavior.

Tangent-based hyper-parameters have previously been explored in data-driven methods primarily as a means of accelerating convergence, and only in settings where instabilities are absent [30, 26, 19]. To the best of the authors' knowledge, the present work is the first to demonstrate that the choice of hyper-parameters, together with an appropriate continuation strategy, plays a critical role in the treatment of material instabilities within a data-driven framework.

The observed behavior can be interpreted in close analogy with finite element formulations employing constitutive models. In such settings, non-convexity arising from material softening is known to cause standard Newton iterations to fail, motivating the use of arc-length solvers and consistent material tangents to robustly trace snap-back responses. In this context, the present data-driven formulation using an arc-length solver together with tangent-based hyper-parameters may be viewed as the analogue of a monolithic Newton method equipped with a consistent material tangent. Also, the regularization method introduced relies on the gradient of the strain variable, and the regularization is removed once the damage variable reaches 1. In [23] the authors analyze the effectiveness of regularization in such cases and conclude that the regularization in such cases is ineffective and the damage evolution is discontinuous. Coupling the method with an arc-length solver, however, forces the damage evolution to be continuous and keeps the regularization effective, as demonstrated.

Having established the algorithmic requirements for handling instabilities, the role of regularization becomes central in ensuring mesh-independent responses. Incorporating strain-based Lipschitz regularization within the data-driven framework yields mesh-independent damage profiles in one and two dimensions. At the same time, the results highlight practical challenges associated with inequality-based regularization, including oscillatory force-displacement responses.

In the two-dimensional setting, the regularization strategy was therefore modified to a log-barrier-type formulation with the objective of suppressing these oscillations. For the parameter values considered in the present study, a saw-tooth response was still observed. Whether the log-barrier reduces the oscillation amplitude relative to the strict-inequality formulation is not tested here. It is nevertheless plausible that alternative choices of the barrier parameter may further smooth the global response, a possibility that was not explored here. Also, as noted in Section 6.1, the implicit database used in the current article does not contain the tension-compression asymmetry that is typically employed in the continuum damage setting to prevent healing. Incorporating such asymmetry into a data-driven database, together with the associated data structures, may present considerable difficulties and is left as an open problem for future work.

The use of inequality- or barrier-based regularization in the present work is motivated by the desire to delay the influence of strain-gradient effects, rather than activating them from the onset of deformation as in classical gradient elasticity formulations such as those of Mindlin or Toupin. In this sense, the regularization acts primarily as a stabilization mechanism that becomes relevant only once localization develops, rather than as a constitutive modification that alters the material response throughout the loading process.

8. Conclusion

This work examined the behavior of DDCM minimization algorithm in the presence of damage-induced softening and snapback instabilities. By analytically defining the material response, database discretization effects have been removed and the algorithmic aspects have been isolated.

The choice of hyper-parameter is shown to govern not only convergence rate but also branch selection under softening; tangent-based hyper-parameters combined with arc-length continuation are required to recover localized solutions. In the absence of regularization, the data-driven algorithm reproduces the expected mesh-dependent localization behavior, provided the algorithmic ingredients are chosen appropriately. A strain-gradient-based regularization strategy was then introduced within the DD setting, yielding mesh-objective localization widths in both one- and two-dimensional problems.

Overall, the algorithmic behavior was determined using an implicit (infinite) database to keep it free of sampling effects. With that resolved, the criteria identified here are expected to carry over to discrete experimental databases, modulo database-specific effects (sampling and phase-space coverage) whose systematic study is left as future work.

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During the preparation of this work, the author(s) used Claude in order to improve the readability of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

A. Material update algorithm

Algorithms 2 and 3 give the per-Gauss-point material update used from Section 3.2 onward, in the form it is implemented. Algorithm 2 performs the branch selection described in Section 3.3.1: it computes closest points on the elastic and softening branches, checks each against the admissibility constraints of Definition 1, and returns the admissible material state together with its tangent for use as the element-wise hyper-parameter $\mathbf{C}$. The closest-point search on the softening branch is nonlinear and is solved by Newton iteration; Algorithm 3 assembles the residual (55) and Hessian (56) of Appendix B for a single Newton step.

Algorithm 2 Material update with branch admissibility (per-Gauss-point)

Require: Mechanical state (σ_i, ϵ_i); previous damage D^n; previous projection ϵ^{*n}; history $\mathcal{H}^n$; hyper-parameter $\mathbf{C}$; $\mathbf{H}_0$; (k, Y_c); cap $d_{\max}$; tolerance η; iteration cap $N_{\max}$
Ensure: $(\epsilon^*{}_{i+1}, \sigma^*{}_{i+1}, \partial\sigma^*/\partial\epsilon^*,\ D^{n+1})$

Part A — Closest point on the elastic (degraded) branch

$g_n \leftarrow g(D^n)$ via eq. (23); $\mathbf{C}_e \leftarrow g_n\,\mathbf{H}_0$
$(\mathbf{H}_e, \mathbf{r}_e) \leftarrow$ ASSEMBLE$(\epsilon_i, \sigma_i, \epsilon_i, \mathbf{C}_e, D^n;$ ELASTIC$)$
$\epsilon^*{}_e \leftarrow \epsilon_i - \mathbf{H}_e^{-1}\mathbf{r}_e$ ▷ closed-form analogue of eq. (50) via one Newton step
$\sigma^*{}_e \leftarrow \mathbf{C}_e\,\epsilon^*{}_e$; $\psi_0^e \leftarrow \frac{1}{2}\,\epsilon^*{}_e{\cdot}\mathbf{H}_0\,\epsilon^*{}_e$; $Y_e \leftarrow -g'(D^n)\,\psi_0^e$

Part B — Closest point on the damage branch

$\epsilon^*{}_d \leftarrow \epsilon^{*n}$; $\mathbf{C}_d \leftarrow \mathbf{C}$; $\text{state}_d^{\text{conv}} \leftarrow$ `True`
$(\mathbf{H}_d, \mathbf{r}_d, \mathbf{T}_d) \leftarrow$ ASSEMBLE$(\epsilon^*{}_d, \sigma_i, \epsilon_i, \mathbf{C}_d, D^n;$ DAMAGE$)$ ▷ residual eq. (55), Hessian eq. (56)
$\mathbf{C}_d \leftarrow \mathbf{T}_d$; re-assemble $(\mathbf{H}_d, \mathbf{r}_d, \mathbf{T}_d)$
for $j = 1, \dots, N_{\max}$ **do**
  $\Delta\epsilon^* \leftarrow -\mathbf{H}_d^{-1}\mathbf{r}_d$; $\epsilon^*{}_d \leftarrow \epsilon^*{}_d + \Delta\epsilon^*$
  Re-assemble $(\mathbf{H}_d, \mathbf{r}_d, \mathbf{T}_d)$ at the new $\epsilon^*{}_d$; $\mathbf{C}_d \leftarrow \mathbf{T}_d$
  if $\|\Delta\epsilon^*\| < \eta$ **then break**
  end if
end for
if $j = N_{\max}$ **then** $\text{state}_d^{\text{conv}} \leftarrow$ `False`
end if
$\psi_0^d \leftarrow \frac{1}{2}\,\epsilon^*{}_d{\cdot}\mathbf{H}_0\,\epsilon^*{}_d$; $d \leftarrow \left(\sqrt{k\psi_0^d} - \sqrt{Y_c}\right)/\left((k-1)\sqrt{Y_c}\right)$; $\sigma^*{}_d \leftarrow g(d)\,\mathbf{H}_0\,\epsilon^*{}_d$

Part C — Full-damage saturation

if $d > d_{\max}$ **then**
  $d \leftarrow d_{\max}$; recompute $(\epsilon^*{}_d, \sigma^*{}_d)$ by one elastic-style Newton step at $g(d_{\max})$
  return $(\epsilon^*{}_d, \sigma^*{}_d, \mathbf{T}_d, D)$
end if

Part D — Admissibility and selection

$\text{state}_e \leftarrow (\psi_0^e < \mathcal{H}^n)$
$\text{state}_d \leftarrow (D > D^n) \wedge \text{state}_d^{\text{conv}}$
$\delta_e \leftarrow 2\,\mathcal{F}(\epsilon^*{}_e; \mathbf{C}_e)$, $\delta_d \leftarrow 2\,\mathcal{F}(\epsilon^*{}_d; \mathbf{C}_d)$
if $\text{state}_e \wedge \text{state}_d$ **then**
  return $(\epsilon^*{}_e, \sigma^*{}_e, \mathbf{C}_e, D^n)$ if $|\delta_e| < |\delta_d|$, else $(\epsilon^*{}_d, \sigma^*{}_d, \mathbf{T}_d, D)$
else if $\text{state}_e \wedge \neg\,\text{state}_d$ **then**
  return $(\epsilon^*{}_e, \sigma^*{}_e, \mathbf{C}_e, D^n)$
else if $\neg\,\text{state}_e \wedge \text{state}_d$ **then**
  return $(\epsilon^*{}_d, \sigma^*{}_d, \mathbf{T}_d, D)$
else
  $\alpha \leftarrow \sqrt{Y_c/Y_e}$
  return $(\alpha\,\epsilon^*{}_e,\ \alpha\,\sigma^*{}_e,\ \mathbf{T}_d,\ D^n)$ ▷ forced projection onto yield surface
end if

Algorithm 3 Residual and Hessian assembly for the material update Newton iteration.

procedure ASSEMBLE($\epsilon^*, \sigma_i, \epsilon_i, \mathbf{C}, D^n$; MODE)
  $\mathbf{S} \leftarrow \mathbf{C}^{-1}$; $\hat{\boldsymbol{\sigma}}_0 \leftarrow \mathbf{H}_0\epsilon^*$; $\psi_0 \leftarrow \frac{1}{2}\,\epsilon^* \cdot \hat{\boldsymbol{\sigma}}_0$
  if MODE=ELASTIC **then**
    $D \leftarrow D^n$
  else
    $D \leftarrow (\sqrt{k\psi_0} - \sqrt{Y_c})/((k-1)\sqrt{Y_c})$, clamped to $[0, 1]$
  end if
  $g \leftarrow g(D)$ via eq. (23); $\sigma^* \leftarrow g\,\mathbf{H}_0\epsilon^*$
  if MODE=ELASTIC **or** D clamped **then**
    $\mathbf{T} \leftarrow g\,\mathbf{H}_0$ ▷ $g'=0$ limit of eq. (53)
  else
    $\mathbf{T} \leftarrow g\,\mathbf{H}_0 - \dfrac{\sqrt{kY_c}}{2(k-1)\,\psi_0^{3/2}}\,\hat{\boldsymbol{\sigma}}_0 \otimes \hat{\boldsymbol{\sigma}}_0$ ▷ eq. (53)
  end if
  $\mathbf{r} \leftarrow -\mathbf{T}\,\mathbf{S}\,(\sigma_i - \sigma^*) - \mathbf{C}\,(\epsilon_i - \epsilon^*)$ ▷ eq. (55)
  if MODE=DAMAGE **and** D not clamped **then**
    $\mathbf{H} \leftarrow \mathcal{T} : \mathbf{S}(\sigma_i - \sigma^*) + \mathbf{T}\,\mathbf{S}\,\mathbf{T} + \mathbf{C}$ ▷ eq. (56)
  else
    $\mathbf{H} \leftarrow \mathbf{T}\,\mathbf{S}\,\mathbf{T} + \mathbf{C}$ ▷ $g'=0$ limit of eq. (56)
  end if
  return $(\mathbf{H}, \mathbf{r}, \mathbf{T})$
end procedure

B. Material update

When the mechanical state is given, the material state can be updated analytically in the present case.

In 1D

In 1D, assuming that the database is represented by a model of the form $\sigma^* = m\epsilon^* + c$, the material state is the minimizer of the following functional:

$$\frac{(\sigma - \sigma^*)^2}{2C} + \frac{C}{2}(\epsilon - \epsilon^*)^2. \tag{48}$$

Taking its derivative with respect to ϵ^* and equating it to zero, the material state can be seen to be

$$\epsilon^* = \frac{m\sigma + C^2\epsilon - mc}{m^2 + C^2}, \quad \sigma^* = m\epsilon^* + c. \tag{49}$$

The point on the elastic branch can be found by simply using $m = E, c = 0$ in the above equation.

$$\epsilon^* = \frac{E\sigma + C^2\epsilon}{E^2 + C^2}, \quad \sigma^* = E\epsilon^*. \tag{50}$$

In 2D

The generating model is ${\sigma^*}_i = g(D)E_{ij}{\epsilon^*}_j$. g is the degradation function of the damage variable, D. The damage evolution in the present case is taken to be a function of strain as (see equation 25)

$$D = \frac{\sqrt{k\psi_0(\boldsymbol{\epsilon}^*)} - \sqrt{Y_c}}{(k-1)\sqrt{Y_c}}.$$

ψ_0 is the strain energy density. Y_c and k are material parameters. g is then defined as (see equation 23)

$$g(D) = \frac{1 - D}{1 + (k-1)D}.$$

The distance functional, written in index form is

$$d(\dots) = \frac{1}{2}(\sigma - \sigma^*)_i S_{ij}(\sigma - \sigma^*)_j + \frac{1}{2}(\epsilon - \epsilon^*)_i C_{ij}(\epsilon - \epsilon^*)_j. \tag{51}$$

The material state can then be found by minimizing the above functional with respect to ϵ^*. Thus, $\dfrac{\partial d}{\partial \epsilon^*} = 0$. This can be further developed as follows:

$$\frac{\partial d}{\partial {\epsilon^*}_k} = -\frac{\partial {\sigma^*}_i}{\partial {\epsilon^*}_k} S_{ij}(\sigma - \sigma^*)_j - \frac{\partial {\epsilon^*}_i}{\partial {\epsilon^*}_k} C_{ij}(\epsilon - \epsilon^*)_j. \tag{52}$$

Realizing that ${\sigma^*}_i = g(d)E_{ij}{\epsilon^*}_j$, the derivative can be written as

$$\frac{\partial {\sigma^*}_i}{\partial {\epsilon^*}_k} = g'(D)\frac{\partial D}{\partial {\epsilon^*}_k}E_{ij}{\epsilon^*}_j + g(D)E_{ij}\delta_{jk}, \tag{53}$$

$$= -\frac{\sqrt{kY_c}}{2(k-1)\psi_0\sqrt{\psi_0}}\hat{\sigma}^*{}_i\hat{\sigma}^*{}_k + g(D)E_{ik}. \tag{54}$$

Thus, the equation to be resolved can be seen to be

$$r_k = \left(\frac{\sqrt{kY_c}}{2(k-1)\psi_0\sqrt{\psi_0}}\hat{\sigma}^*{}_i\hat{\sigma}^*{}_k + g(D)E_{ik}\right)S_{ij}(\sigma-\sigma^*)_j - C_{kj}(\epsilon-\epsilon^*)_j = 0. \tag{55}$$

The above equation is non-linear and requires newton's method to solve. This requires the derivative of the residual, which is given by

$$\frac{\partial r_k}{\partial {\epsilon^*}_p} = \left\{\frac{\sqrt{kY_c}}{2(k-1)}\left[\frac{E_{kp}\hat{\sigma}^*{}_i + \hat{\sigma}^*{}_k E_{ip} + \hat{\sigma}^*{}_p E_{ik}}{\psi_0\sqrt{\psi_0}} - \frac{3}{2}\psi_0^{-5/2}\hat{\sigma}^*{}_i\hat{\sigma}^*{}_k\hat{\sigma}^*{}_p\right]S_{ij}(\sigma-\sigma^*)_j\right\} + \left(\frac{\sqrt{kY_c}}{2(k-1)\psi_0\sqrt{\psi_0}}\hat{\sigma}^*{}_i\hat{\sigma}^*{}_k + g(D)E_{ik}\right)S_{ij}\left(\frac{\sqrt{kY_c}}{2(k-1)\psi_0\sqrt{\psi_0}}\hat{\sigma}^*{}_j\hat{\sigma}^*{}_p + g(D)E_{jp}\right) + C_{kp}. \tag{56}$$

The elastic case can be obtained by taking $g' = 0$ in equation 53.

C. Analysis of convergence

The mechanical and material updates for the 1D case will be described here. Restating the problem, the element in the middle is already taken to lay on the damage branch, while the other elements are unloading on the elastic branch. See Figure 2. The distance functional is

$$d(\sigma, \sigma_1^*, \sigma_2^*, \epsilon_1, \epsilon_2; u_g) = \frac{N\ell_e}{2}\left[\frac{(\sigma-\sigma_1^*)^2}{C_1} + C_1(\epsilon_1-\epsilon_1^*)^2\right] + \frac{\ell_e}{2}\left[\frac{(\sigma-\sigma_2^*)^2}{C_2} + C_2(\epsilon_2-\epsilon_2^*)^2\right].$$

The compatibility condition is $N\epsilon_1 + \epsilon_2 = u_D/\ell_e$. The mechanical state of the bar is described by a scalar stress $\sigma \in \mathbb{R}$ and strains $\epsilon_1, \epsilon_2 \in \mathbb{R}$ (ϵ_1 for the undamaged elements and ϵ_2 for the damaged element). The material state is described by $(\epsilon_1^*, \sigma_1^*) \in \mathbb{R}^{1\times 1}$ and $(\epsilon_2^*, \sigma_2^*) \in \mathbb{R}^{1\times 1}$.

Now, taking the variations with respect to the compatibility equation as a constraint, the system of equations for ϵ_1, ϵ_2 and σ are obtained as

$$\begin{bmatrix}\epsilon_1\\ \epsilon_2\\ \sigma\end{bmatrix} = \begin{bmatrix} C_1/(C_1+NC_2) & -C_2/(C_1+NC_2) & 0 & 0\\ -NC_1/(C_1+NC_2) & NC_2/(C_1+NC_2) & 0 & 0\\ 0 & 0 & \frac{NC_2}{C_1+NC_2} & \frac{C_1}{C_1+NC_2}\end{bmatrix}\begin{bmatrix}\epsilon_1^*\\ \epsilon_2^*\\ \sigma_1^*\\ \sigma_2^*\end{bmatrix} + \begin{bmatrix} C_2u_D/(C_1+NC_2)/\ell_e\\ C_1u_D/(C_1+NC_2)/\ell_e\\ 0\end{bmatrix}. \tag{57}$$

The material states can be found by taking the variations with respect to $\epsilon_1^*, \epsilon_2^*$, while realizing that $\sigma_1^* = E\epsilon_1^*$, and $\sigma_2^* = m\epsilon_2^* + c$. Thus,

$$\begin{bmatrix}\epsilon_1^*\\ \epsilon_2^*\\ \sigma_1^*\\ \sigma_2^*\end{bmatrix} = \begin{bmatrix} C_1^2/(E^2+C_1^2) & 0 & E/(E^2+C_1^2)\\ 0 & C_2^2/(m^2+C_2^2) & m/(m^2+C_2^2)\\ EC_1^2/(E^2+C_1^2) & 0 & E^2/(E^2+C_1^2)\\ 0 & mC_2^2/(m^2+C_2^2) & m^2/(m^2+C_2^2)\end{bmatrix}\begin{bmatrix}\epsilon_1\\ \epsilon_2\\ \sigma\end{bmatrix} + \begin{bmatrix} 0\\ -mc/(m^2+C_2^2)\\ 0\\ cC_2^2/(m^2+C_2^2)\end{bmatrix}. \tag{58}$$

The system of equations corresponding to the mechanical and material updates can be written in short as $\boldsymbol{a} = \boldsymbol{A}\boldsymbol{b} + \boldsymbol{c}$, and $\boldsymbol{b} = \boldsymbol{C}\boldsymbol{a} + \boldsymbol{d}$. The alternating minimization can then be written as

$$\boldsymbol{b}^{n+1} = \boldsymbol{C}\boldsymbol{a}^n + \boldsymbol{d},$$

$$\boldsymbol{a}^{n+1} = \boldsymbol{A}\boldsymbol{b}^{n+1} + \boldsymbol{c}.$$

Combining them together,

$$\boldsymbol{a}^{n+1} = \boldsymbol{AC}\boldsymbol{a}^{n} + \boldsymbol{Ad} + \boldsymbol{c}. \tag{59}$$

At convergence, $\boldsymbol{a}^{n+1} \approx \boldsymbol{a}^{n} \approx \boldsymbol{a}$. Thus, $\boldsymbol{e}^{n+1} = \boldsymbol{AC}\boldsymbol{e}^{n}$, where $\boldsymbol{e}^{n+1} = \boldsymbol{a}^{n+1} - \boldsymbol{a}$ is the error at the iteration $n+1$. The rate of convergence of the method depends on the spectral radius of the matrix $\boldsymbol{AC}$, which in turn depends on the metric chosen.

With arc length solver

Snapback occurs when the mesh and thus, the size of the damage zone, is refined. In this case, an arc length solver is needed to track the full damage evolution in the element. So, this case will first be presented and the effect of metric on the rate of convergence will be studied.

Using an arc length solver corresponds to when the displacement at the end of the bar, u_D, is treated as an unknown. This results in the number of unknowns increasing by 1. The additional equation that is needed to fully resolve the system is the arc length constraint, stated in equation 62 is $\epsilon_2 = \epsilon_0 + \Delta\epsilon$. This changes the equations to be resolved in the previous section to

$$\begin{bmatrix} 1 & 0 & 0 & -C_2/(C_1+NC_2)/\ell_e \\ 0 & 1 & 0 & -C_1/(C_1+NC_2)/\ell_e \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \sigma \\ u_D \end{bmatrix} = \begin{bmatrix} C_1/(C_1+NC_2) & -C_2/(C_1+NC_2) & 0 & 0 \\ -NC_1/(C_1+NC_2) & NC_2/(C_1+NC_2) & 0 & 0 \\ 0 & 0 & \frac{NC_2}{C_1+NC_2} & \frac{C_1}{C_1+NC_2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \epsilon_1^* \\ \epsilon_2^* \\ \sigma_1^* \\ \sigma_2^* \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \epsilon_0 + \Delta\epsilon \end{bmatrix}. \tag{60}$$

The material updates remain unchanged and can be found by taking the variations with respect to $\epsilon_1^*, \epsilon_2^*$, while realizing that $\sigma_1^* = E\epsilon_1^*$, and $\sigma_2^* = m\epsilon_2^* + c$. Thus,

$$\begin{bmatrix} \epsilon_1^* \\ \epsilon_2^* \\ \sigma_1^* \\ \sigma_2^* \end{bmatrix} = \begin{bmatrix} C_1^2/(E^2+C_1^2) & 0 & E/(E^2+C_1^2) & 0 \\ 0 & C_2^2/(m^2+C_2^2) & m/(m^2+C_2^2) & 0 \\ EC_1^2/(E^2+C_1^2) & 0 & E^2/(E^2+C_1^2) & 0 \\ 0 & mC_2^2/(m^2+C_2^2) & m^2/(m^2+C_2^2) & 0 \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \sigma \\ u_D \end{bmatrix} + \begin{bmatrix} 0 \\ -mc/(m^2+C_2^2) \\ 0 \\ cC_2^2/(m^2+C_2^2) \end{bmatrix}. \tag{61}$$

Thus, even in this case, the update can be cast into the form of equation 59 and can be analyzed similarly.

D. Analysis with a different starting point

The analyses of section 3.1 corresponding to cases 2 and 3 have been repeated using a different starting point that is further down the damage branch. That is, $\sigma = \sigma_1^* = \sigma_2^* = 50\sigma_0, \epsilon_1 = \epsilon_1^* = 0.5, \epsilon_2 = \epsilon_2^* = 7.99$. $N = 100$.

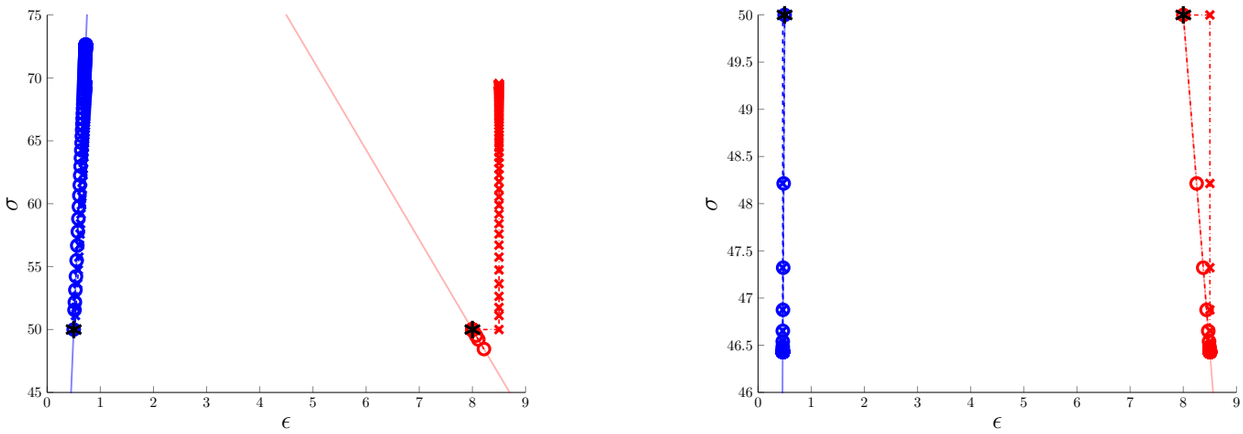
Arc-length control with snapback

With arc-length control and in the presence of a snapback, the trajectories of the mechanical and the material states for the elements on the elastic and the damage branch with $C_2 = g(d)E$ (the degraded modulus) and $C_2 = m$ have been presented in Figures 21a and 21b.

The trajectory is similar to that presented in Figures 5a and 5b. The choice $C_2 = g(d)E > 0$ violates the bound of equation (22), so $\rho(\boldsymbol{AC}) > 1$. The irreversibility constraint 1 becomes active and the iterations stagnate after moving apart initially. With $C_2 = m$, the mechanical and the material states almost coincide after 7 iterations.

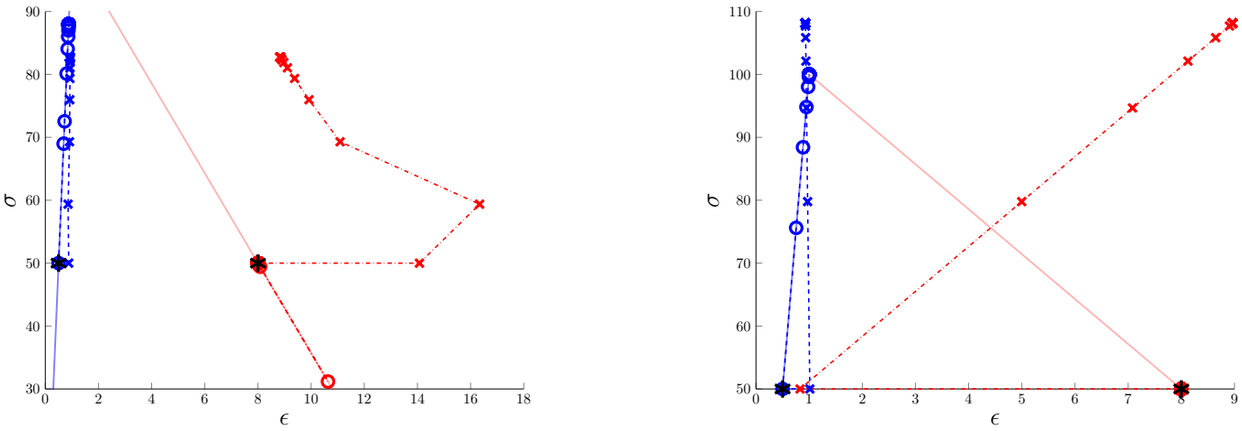
Displacement control with snapback

Under a displacement control and in the presence of a snapback, the evolution of distance is presented in Figure 6. For both $C_2 = g(d)E$ (the secant modulus) and $C_2 = m$, the distance stagnates after a few iterations. The corresponding trajectories of the



(a) (b)

Figure 21: Results from the 1D analysis with a different starting point. Successive mechanical and material states. Case 2: Snapback and arc-length constraint included. (a) $C_2 = g(d)E$. (b) $C_2 = m$.



(a) (b)

Figure 22: Results from the 1D analysis with a different starting point. Successive mechanical and material states. Case 3: Displacement control under snapback. (a) $C_2 = g(d)E$. (b) $C_2 = m$.

mechanical and the material states for the elements on the elastic and the damage branch are presented in Figures 22a and 22b. The trajectory is similar to that presented in Figures 7a and 7b. Both choices for C_2 result in iterates that move further apart and stagnate. The irreversibility constraint 1 is active in both the scenarios while the peak stress constraint 2 becomes active when $C_2 = m$, as presented in Figure 22b.

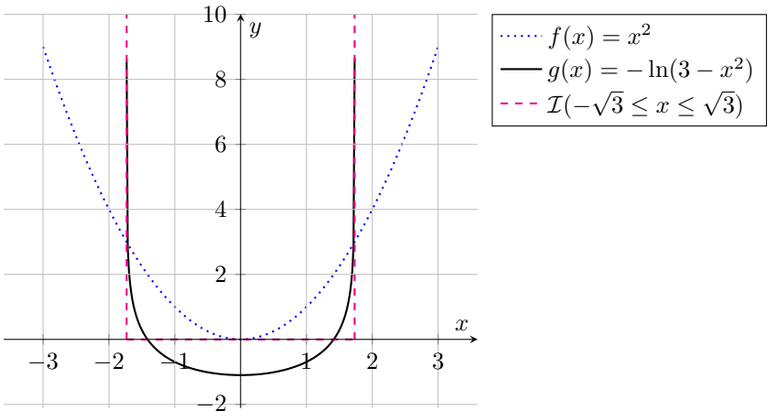
E. Arc length constraint

The arc length constraint is simpler in 1D case than in 2D case as it involves selecting degrees of freedom where a certain quantity is monotone increasing. For instance, in [4, 33], the damage variables are used to write the arc length constraint. In the current case, for the 1D bar, the mechanical strain variable at the left end of the bar is chosen to impose the arc length constraint. The constraint can be expressed as

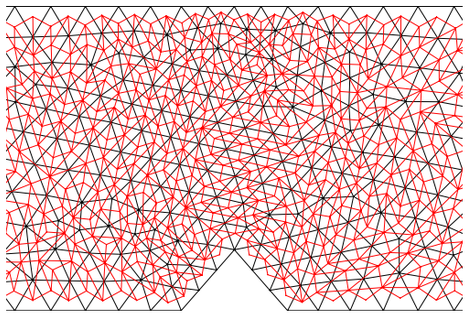
$$\epsilon^{n+1} - \epsilon^n = \Delta, \tag{62}$$

where $n+1$ and n denote the load steps. Δ is the arc length, which in the current case is the strain increment for the element at the extreme left between two steps. Expressing the displacements as $\boldsymbol{u} = \boldsymbol{C}_f \boldsymbol{u}_f + \lambda \boldsymbol{u}_g$, where $\boldsymbol{u}_f$ denotes the unknown/free degrees of freedom and λ, in the current case, is the displacement applied at the right end with u_g as the corresponding degree of freedom. For instance, if the bar contains N nodes numbered 1 to N, $\boldsymbol{u}_f$ contains the DOFs of the nodes 2 to $N-1$. $\boldsymbol{C}_f \in \mathbb{R}^{N \times N-2}$ and $\boldsymbol{u}_g \in \mathbb{R}^N = \{0, 0, 0, \dots 1\}^T$.

In 2D case, the method is slightly involved. The aim is to identify the degrees of freedom where a specific quantity is always increasing. For this purpose, the elements whose material states lie on the 'damage branch' are first identified. The damage levels in these elements are thus increasing in the load step under consideration. Then, an equivalent strain energy density of the mechanical states corresponding to these elements are computed using the mechanical strains. The arc length constraint is now written using



(a) Log barrier function vs quadratic functional



(b) A zoom-in of the displacement mesh (in black) and the associated lip-mesh (in red)

Figure 23: Regularization functions comparison and the displacement, lip meshes for the notched bar.

these degrees of freedom and their equivalent strain energy density. Let the damage-driving energy over an element e be

$$\psi_0^{(e)} = \tfrac{1}{2}\,\boldsymbol{\varepsilon}^{(e)T}\mathbf{C}\boldsymbol{\varepsilon}^{(e)}. \tag{63}$$

The damage evolution term is defined as

$$d^{(e)} = \frac{\sqrt{k\,\psi_0^{(e)}} - \sqrt{Y_c}}{\sqrt{Y_c}(k-1)}. \tag{64}$$

The arc-length constraint is now written as

$$r_{diss} = \sum_e A_e \left[\frac{\sqrt{k\,\psi_0^{(e)}} - \sqrt{Y_c}}{\sqrt{Y_c}(k-1)} - d_{\text{old}}^{(e)} \right] - \Delta = 0. \tag{65}$$

This equation is added to the system of equations 42 - 47 and solved using the Newton method.

F. Discrete version of the constraint

F.1. Constraint imposition

To impose the constraint on the strain variable, the lip mesh is defined along with the FE mesh as in [9]. To compute the derivative of the strain variable, in discrete formulation, the lip mesh is used. The strain variables are computed from the displacements as $\boldsymbol{\epsilon} = \boldsymbol{B}\boldsymbol{u}$, where $\boldsymbol{B}$ is the discrete gradient operator, computed on the FE mesh. Defining the discrete gradient operator on the lip mesh and representing it in a matrix form as $\boldsymbol{G}$, the gradients of strain can be expressed in terms of the displacements as

$$\boldsymbol{k} = \nabla\boldsymbol{\epsilon} = \boldsymbol{B}'\boldsymbol{B}\boldsymbol{u} = \boldsymbol{G}\boldsymbol{u}, \tag{66}$$

where, $\boldsymbol{k}$ is the matrix containing the gradients of strain as $\begin{bmatrix} k_{111} & k_{211} & k_{122} & k_{222} & k_{112} & k_{212} \end{bmatrix}^T$ and $\boldsymbol{B}'$ denotes the discrete gradient operator defined on the lip-mesh.

The constraint on the strain gradient can thus be written as

$$(\boldsymbol{G}\boldsymbol{u})^T\boldsymbol{G}\boldsymbol{u} \leq \frac{1}{\ell_c^2}, \tag{67}$$

Figure 23a compares the energy responses of different terms used to enforce the strain-gradient constraint. The standard quadratic form increases rapidly for small deviations from zero, leading to early activation near the origin. In contrast, the log-barrier-type term remains nearly flat in a neighborhood of zero, thereby postponing the onset of localization until the strain gradients begin to concentrate.

F.2. Stability check

The stability of the discrete displacement-strain pairing is a special case of the Ladyzhenskaya-Babuska-Brezzi (LBB) condition for mixed finite element methods; see [2] for the general theory. The stability of the displacement-strain combination used in this article will now be studied.

Considering that the total energy can be expressed as a sum of quadratic energies of the strain and the strain gradient terms, the solution is the collection that renders the Lagrangian stationary. In discrete setting, using $\boldsymbol{u} \in \mathbb{R}^N$ for displacement, $\boldsymbol{Bu}$ for symmetric gradient of the displacement, piecewise constant over the displacement mesh, and $\boldsymbol{\epsilon} \in \mathbb{R}^M$ for strain, piecewise affine on the lip-mesh, the Lagrangian is

$$\Pi = \frac{1}{2}\boldsymbol{u}^T \boldsymbol{K}\boldsymbol{u} + \frac{1}{2}\boldsymbol{\epsilon}^T \boldsymbol{L}\boldsymbol{\epsilon} + \boldsymbol{\Lambda}^{\mathbf{T}} (\boldsymbol{\epsilon} - \boldsymbol{Bu}), \tag{68}$$

where $\boldsymbol{K}, \boldsymbol{L}$ denote the stiffness matrix corresponding to strain-energy and strain gradient-energy, respectively. $\boldsymbol{\Lambda} \in \mathbb{R}^M$ denotes the array of Lagrange multipliers that enforce the relation between the strain $\boldsymbol{\epsilon}$ and the symmetric gradient of displacement $\boldsymbol{Bu}$. The values of $\boldsymbol{Bu}$ at the centroids of displacement mesh are used to evaluate the nodal values of $\boldsymbol{\epsilon}$ on the strain mesh. The discrete form $\boldsymbol{\Lambda}^{\mathbf{T}} (\boldsymbol{\epsilon} - \boldsymbol{Bu})$ can thus be used to define its continuous counterpart as

$$\int \lambda \left(\epsilon - \text{sym grad}(u)\right) \mathrm{d}\Omega. \tag{69}$$

Using $\lambda = \sum_{i=1}^{M} \Lambda_i \delta(x - x_i)$, with x_i denoting the coordinates of the centroids of the displacement mesh, the discrete version of the constraint term can be obtained.

After taking variations of Π, the bilinear term associated with constraints can be identified as

$$b^h(\boldsymbol{U}, \boldsymbol{\Lambda}) = \boldsymbol{\Lambda}^{\mathbf{T}} (\boldsymbol{\epsilon} - \boldsymbol{Bu}), \tag{70}$$

where $\boldsymbol{U} = [\boldsymbol{u}; \boldsymbol{\epsilon}]^T$. Considering the term

$$\frac{b^h(\boldsymbol{U}, \boldsymbol{\Lambda})}{\|\boldsymbol{U}\| \|\boldsymbol{\Lambda}\|},$$

with $\|\boldsymbol{U}\|$ denoting the norm of the vector $[\boldsymbol{u}; \boldsymbol{\epsilon}]^T$, and evaluating it when $\boldsymbol{u} = \boldsymbol{0}$, $\boldsymbol{\epsilon} = \boldsymbol{\Lambda}$,

$$\frac{b^h(\boldsymbol{U}, \boldsymbol{\Lambda})}{\|\boldsymbol{U}\| \|\boldsymbol{\Lambda}\|} = \frac{\boldsymbol{\Lambda}^{\mathbf{T}} \boldsymbol{\Lambda}}{\|\boldsymbol{\Lambda}\|^{\mathbf{2}}} = 1. \tag{71}$$

Thus,

$$\sup_{\boldsymbol{U}} \frac{b^h(\boldsymbol{U}, \boldsymbol{\Lambda})}{\|\boldsymbol{U}\| \|\boldsymbol{\Lambda}\|} \geq 1, \tag{72}$$

and so,

$$\beta = \inf_{\boldsymbol{\Lambda}} \sup_{\boldsymbol{U}} \frac{b^h(\boldsymbol{U}, \boldsymbol{\Lambda})}{\|\boldsymbol{U}\| \|\boldsymbol{\Lambda}\|} \geq 1. \tag{73}$$

Thus, the displacement-strain combination satisfies the stability requirements independently of the mesh size, with a constant $\beta \geq 1$.

The LBB condition is in general nontrivial because it requires mesh-independent invertibility of the constraint operator. In the present setting, however, the constraint takes the form

$$\mathbf{I}\,\boldsymbol{\epsilon} - \mathbf{B}\mathbf{u} = \mathbf{0},$$

where $\mathbf{I}$ is the identity on the strain degrees of freedom – a direct point-by-point evaluation at the lip-mesh nodes rather than an operator that must itself be inverted. For every displacement $\mathbf{u}$, this fixes a unique strain $\boldsymbol{\epsilon}$, so injectivity of the constraint operator is guaranteed trivially and the inf-sup bound follows from a direct substitution. A formal functional-space treatment is beyond the scope of this article.

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